United States Patent	12415172
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Miansarigavzan; Morteza et al.

Systems and method for automated oligonucleotide synthesis

Abstract

A reactor system is disclosed. The reactor system includes a vessel configured to contain a solid support, the vessel including: a vessel wall defining a reaction chamber, the reaction chamber having a first end and a second end opposite the first end; a piston operatively arranged at the first end and configured to translate within the reaction chamber; a force measuring device coupled to the piston and configured to measure a load on the piston; a piston driver coupled to the piston; and a processor operably coupled to the force measuring device and the piston driver, the processor configured to: measure a load on the piston using the force measuring device, and adjust a position of the piston using the piston driver based on measuring the load on the piston.

Inventors: **Miansarigavzan; Morteza (Redwood City, CA), Datwani; Sammy (Redwood City, CA)**

Applicant: **Synthego Corporation** (Redwood City, CA)

Family ID: **1000008818534**

Assignee: **Synthego Corporation (Redwood City, CA)**

Appl. No.: **18/544146**

Filed: **December 18, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240207804 A1	Jun. 27, 2024

Related U.S. Application Data

us-provisional-application US 63445536 20230214
us-provisional-application US 63434777 20221222

Publication Classification

Int. Cl.: **B01J19/18** (20060101); **B01J19/00** (20060101)

U.S. Cl.:

CPC **B01J19/0046** (20130101); **B01J19/18** (20130101); B01J2219/0029 (20130101); B01J2219/00353 (20130101); B01J2219/00477 (20130101); B01J2219/00596 (20130101); B01J2219/00686 (20130101); B01J2219/00698 (20130101); B01J2219/00722 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2677271	12/1953	Faris, Jr.	73/789	G01N 3/20
4121289	12/1977	Stiel	702/41	A61J 3/10
4299794	12/1980	Kelley	N/A	N/A
4680958	12/1986	Ruelle	73/54.14	G01N 11/04
4728502	12/1987	Hamill	N/A	N/A
4800166	12/1988	Horn	N/A	N/A
4861866	12/1988	Durrum	530/333	C07K 1/045
5112575	12/1991	Whitehouse	N/A	N/A
5190075	12/1992	Tentler	N/A	N/A
5273715	12/1992	Bridgham	N/A	N/A
5324426	12/1993	Joseph	210/656	B01D 15/14
5411697	12/1994	McGraw	428/903.3	B30B 9/3007
5460786	12/1994	Nokihara	422/138	B01J 19/0046
5681474	12/1996	Gunther	210/91	G01N 30/6021
5807525	12/1997	Allen	N/A	N/A
5851494	12/1997	Dillard	422/50	B01J 19/0046
6028172	12/1999	Stepaniuk	N/A	N/A
6036923	12/1999	Laugharn	N/A	N/A
6168914	12/2000	Campbell	N/A	N/A
6221332	12/2000	Thumm	N/A	N/A
6248521	12/2000	Van Ness	N/A	N/A
6335438	12/2001	Fonnum	N/A	N/A
6395235	12/2001	Kilcoin	N/A	N/A
6409925	12/2001	Gombinsky	N/A	N/A
6416642	12/2001	Alajoki	N/A	N/A
6537817	12/2002	Papen	N/A	N/A

6544775	12/2002	Brugger	N/A	N/A
6566494	12/2002	Jensen	N/A	N/A
6569672	12/2002	Laugharn	N/A	N/A
6613580	12/2002	Chow	N/A	N/A
6685884	12/2003	Stylli	N/A	N/A
6772642	12/2003	Hajduk	N/A	N/A
6858186	12/2004	Hilson	N/A	N/A
6887990	12/2004	Bhan	N/A	N/A
6899137	12/2004	Unger	N/A	N/A
6915713	12/2004	Kipke	N/A	N/A
6929030	12/2004	Unger	N/A	N/A
6949176	12/2004	Vacca	N/A	N/A
7013709	12/2005	Hajduk	N/A	N/A
7052545	12/2005	Unger	N/A	N/A
7101515	12/2005	Schwalbe	N/A	N/A
7132053	12/2005	Hauck	210/656	B01D 15/20
7150994	12/2005	Bergh	N/A	N/A
7160511	12/2006	Takahashi	N/A	N/A
7195670	12/2006	Hansen	N/A	N/A
7216671	12/2006	Unger	N/A	N/A
7223363	12/2006	McNeely	N/A	N/A
7250128	12/2006	Unger	N/A	N/A
7306672	12/2006	Hansen	N/A	N/A
7326296	12/2007	Quake	N/A	N/A
7431685	12/2007	Frey	N/A	N/A
7431837	12/2007	Cohee	N/A	N/A
7431884	12/2007	Ryan	N/A	N/A
7459022	12/2007	Hansen	N/A	N/A
7479186	12/2008	Quake	N/A	N/A
7517440	12/2008	Anex	N/A	N/A
7569354	12/2008	Okano	N/A	N/A
7644759	12/2009	Davidson	N/A	N/A
7670429	12/2009	Quake	N/A	N/A
7678335	12/2009	Kreuzer	N/A	N/A
7704322	12/2009	Hansen	N/A	N/A
7754010	12/2009	Unger	N/A	N/A
7763685	12/2009	Michielin	N/A	N/A
7766055	12/2009	Unger	N/A	N/A
7846392	12/2009	Chow	N/A	N/A
7867757	12/2010	Karlsen	N/A	N/A
7901939	12/2010	Ismagliov	N/A	N/A
7968060	12/2010	Tuyl	N/A	N/A
8021480	12/2010	Hansen	N/A	N/A
8104497	12/2011	Unger	N/A	N/A
8263387	12/2011	Pagano	N/A	N/A
8372340	12/2012	Bird	N/A	N/A
8382896	12/2012	Hansen	N/A	N/A
8414774	12/2012	LaMarr	N/A	N/A
8486340	12/2012	Lee	N/A	N/A
8491178	12/2012	Breidenthal	N/A	N/A

8500852	12/2012	Galbraith	N/A	N/A
8546561	12/2012	Kilambi	N/A	N/A
8550119	12/2012	Unger	N/A	N/A
8592498	12/2012	Enomura	N/A	N/A
8597592	12/2012	Wang	N/A	N/A
8656958	12/2013	Unger	N/A	N/A
8709152	12/2013	Hansen	N/A	N/A
8784745	12/2013	Nelson	N/A	N/A
8791176	12/2013	Van De Runstraat	N/A	N/A
8920752	12/2013	Tisone	N/A	N/A
9034272	12/2014	Cedillo	N/A	N/A
9069358	12/2014	Demmitt	N/A	N/A
9221921	12/2014	Fourarge	N/A	N/A
9273308	12/2015	Link	N/A	N/A
9519000	12/2015	Wilson	N/A	N/A
9522380	12/2015	Herman	N/A	N/A
9545737	12/2016	Henn	N/A	N/A
9555342	12/2016	Wheeler	N/A	N/A
9643136	12/2016	Hansen	N/A	N/A
9651568	12/2016	Putnam	N/A	N/A
9670451	12/2016	Dilorenzo	N/A	N/A
9695214	12/2016	Simon	N/A	N/A
9764060	12/2016	Nazhat	N/A	N/A
9921166	12/2017	Glezer	N/A	N/A
9931636	12/2017	Ching	N/A	N/A
9968902	12/2017	Chu	N/A	N/A
9993818	12/2017	Taylor	N/A	N/A
10105705	12/2017	Chen	N/A	N/A
10155250	12/2017	Chou	N/A	N/A
10227377	12/2018	Avallin	N/A	N/A
10731181	12/2019	Chen	N/A	N/A
10955067	12/2020	Laser	N/A	N/A
11014060	12/2020	Ledwon	N/A	N/A
11071964	12/2020	Garcia	N/A	N/A
11191723	12/2020	Panagiotou	N/A	N/A
11318440	12/2021	Guthrie	N/A	N/A
11406980	12/2021	Blainey	N/A	N/A
11439970	12/2021	Nguyen	N/A	N/A
11484858	12/2021	Slager	N/A	N/A
11610652	12/2022	Aykol	N/A	N/A
11623201	12/2022	Ramezani	N/A	N/A
11633736	12/2022	Cauley	N/A	N/A
11685889	12/2022	Garst	N/A	N/A
11708553	12/2022	Govind	N/A	N/A
11740255	12/2022	Foley	N/A	N/A
11766655	12/2022	Ledwon	N/A	N/A
11801485	12/2022	Doering	N/A	N/A
2001/0043883	12/2000	Chalk	N/A	N/A
2002/0100714	12/2001	Staats	N/A	N/A

2002/0187564	12/2001	Chow	N/A	N/A
2003/0086829	12/2002	Livesay	N/A	N/A
2004/0181343	12/2003	Wigstrom	N/A	N/A
2005/0112674	12/2004	Palti	N/A	N/A
2005/0158213	12/2004	Tsudome	N/A	N/A
2005/0169816	12/2004	Kirshner	N/A	N/A
2005/0272144	12/2004	Sando	N/A	N/A
2007/0169686	12/2006	Quake	N/A	N/A
2007/0258862	12/2006	Vann	N/A	N/A
2008/0177054	12/2007	Evans	N/A	N/A
2008/0277007	12/2007	Unger	N/A	N/A
2009/0053732	12/2008	Vermesh	N/A	N/A
2009/0291505	12/2008	Sarofim	N/A	N/A
2010/0184232	12/2009	Jepsen	N/A	N/A
2011/0259831	12/2010	Brandt	N/A	N/A
2012/0047668	12/2011	David	N/A	N/A
2012/0135404	12/2011	Schwoebel et al.	N/A	N/A
2012/0171090	12/2011	Chang	N/A	N/A
2012/0315211	12/2011	Mahamuni	N/A	N/A
2014/0097353	12/2013	Ben-Hur	N/A	N/A
2015/0276089	12/2014	Unger	N/A	N/A
2016/0244547	12/2015	Jagannathan	N/A	N/A
2019/0314777	12/2018	Li	N/A	N/A
2019/0366293	12/2018	Banyai et al.	N/A	N/A
2020/0222905	12/2019	Hill	N/A	N/A
2020/0254410	12/2019	Inaka	N/A	N/A
2021/0023514	12/2020	Ramsay	N/A	N/A
2021/0039106	12/2020	Eldridge	N/A	N/A
2021/0061844	12/2020	Slavazza	N/A	N/A
2021/0155476	12/2020	Futerman	N/A	N/A
2021/0187468	12/2020	Gilmore	N/A	N/A
2021/0214664	12/2020	Rao	N/A	N/A
2022/0064208	12/2021	Kreuzer	N/A	N/A
2022/0090183	12/2021	Hill	N/A	N/A
2022/0118455	12/2021	Jebrail	N/A	N/A
2022/0307079	12/2021	Hodko	N/A	N/A
2022/0339598	12/2021	Mueller	N/A	N/A
2022/0362770	12/2021	Blainey	N/A	N/A
2022/0372466	12/2021	Xie	N/A	N/A
2022/0380718	12/2021	Komuckzi	N/A	N/A
2023/0026870	12/2022	Seeberger	N/A	N/A
2023/0041595	12/2022	Wu	N/A	N/A
2023/0083189	12/2022	Burgess	N/A	N/A
2023/0135860	12/2022	Sillard	N/A	N/A
2023/0149887	12/2022	Rogers	N/A	N/A
2023/0158468	12/2022	Johnson	N/A	N/A
2023/0219043	12/2022	Link	N/A	N/A
2023/0257658	12/2022	Hoffman	N/A	N/A
2023/0271155	12/2022	Bickmore	N/A	N/A
2023/0279512	12/2022	Masters	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201579035	12/2009	CN	N/A
205216837	12/2015	CN	N/A
0933125	12/1997	EP	N/A
0386238	12/1998	EP	N/A
0934113	12/2001	EP	N/A
937096	12/2003	EP	N/A
1485204	12/2005	EP	N/A
1324829	12/2006	EP	N/A
1345551	12/2008	EP	N/A
1718411	12/2009	EP	N/A
1392484	12/2012	EP	N/A
3341112	12/2017	EP	N/A
2402460	12/2018	EP	N/A
3749740	12/2019	EP	N/A
4208286	12/2022	EP	N/A
4217112	12/2022	EP	N/A
4251326	12/2022	EP	N/A
2561982	12/2017	GB	N/A
1650240	12/1990	SU	N/A
2000070353	12/1999	WO	N/A
2001019518	12/2000	WO	N/A
2004064999	12/2003	WO	N/A
2004104228	12/2003	WO	N/A
2006043642	12/2005	WO	N/A
2008024319	12/2007	WO	N/A
2010141361	12/2009	WO	N/A
2017023380	12/2016	WO	N/A
2018060354	12/2017	WO	N/A
2018197421	12/2017	WO	N/A
2019035998	12/2018	WO	N/A
2021138210	12/2020	WO	N/A
2021148818	12/2020	WO	N/A
2021188032	12/2020	WO	N/A
2022068937	12/2021	WO	N/A
WO-02095390	12/2021	WO	B01D 15/206
2023004035	12/2022	WO	N/A
2023035003	12/2022	WO	N/A
2023044569	12/2022	WO	N/A

OTHER PUBLICATIONS

“Asahi Axial Compression Synthesis Columns” AsahiKASEI BioProcess. webpage <<https://fluidmgmt.ak-bio.com/products/asahi-acs-column/>> Accessed Mar. 15, 2024. cited by applicant

“Cytiva: Adjustable Oligo Column”, webpage <<cytivalifesciences.com/en/us/shop/oligonucleotide-synthesis/columns/adjustable-oligo-column-for-oligopilot-oligonucleotide-synthesizers-p-03868>> Accessed Mar. 15, 2024. cited by applicant

"Cytiva: Columns" webpage <<https://www.cytivalifesciences.com/en/us/shop/oligonucleotide-synthesis/columns?sort=NameAsc&chunk=1>> Accessed Mar. 15, 2024. cited by applicant
"Galak Chromatography: Glass Chromatography Columns," webpage <<https://galaklc.com/compression-glass-column/>> Accessed Mar. 18, 2024. cited by applicant
"Vapourtec: Precision Flow Chemistry: Variable Bed Flow Reactor (VBFR)" webpage <<https://.vapourtec.com/products/flow-reactors/variable-bed-flow-reactor-features/>> Accessed Mar. 15, 2024. cited by applicant
"YMC Europe GMBH: YMC-Actus: Semi-preparative HPLC columns." webpage <<https://ymc.eu/ymc-actus.html>> Accessed Mar. 15, 2024. cited by applicant
"YMC Home: Preparative LC systems: ECO Plus series (Glass column)" webpage <<https://.ymc.co.jp/en/chromato/glass/eco_plus_glass_columns.html>> Accessed Mar. 15, 2024. cited by applicant
United Kingdom Intellectual Property Office, "Patents Act 1977: Search Report under Section 17" issued for Application No. GB1806565.6, dated Jul. 24, 2018. 2 pages. cited by applicant
International Searching Authority, International Search Report and Written Opinion for International Patent Application No. PCT/US2023/084644, dated Apr. 23, 2024. cited by applicant

---

Primary Examiner: Leung; Jennifer A

Attorney, Agent or Firm: Quarles & Brady, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims the benefit of priority to U.S. Provisional Application No. 63/434,777, filed Dec. 22, 2022, and to U.S. Provisional Application No. 63/445,536, filed Feb. 14, 2023, the contents of each of which are incorporated herein by reference in their entirety.

BACKGROUND
(1) Synthesis of oligonucleotides such as RNA from a solid support requires a series of steps in a synthesis cycle with one nucleotide added to the growing oligonucleotide chain every cycle, as shown in diagram **100** of FIG. **1**. The cycle begins with deprotecting a protected base attached to a solid support through the removal of a trityl group by the addition of an acid such as dichloroacetic acid. Once deprotected, the oligonucleotide is coupled to an amidite corresponding to the base to be added to the growing RNA chain. The resultant oligonucleotide then undergoes oxidation or sulfonation to stabilize the RNA backbone, followed by a capping reaction where each oligonucleotide that does not receive an amidite during coupling receives instead an acetyl cap. Oligonucleotides that successfully add the amidite are then deprotected, and the cycles starts over again.
(2) Small-scale synthesis of short oligonucleotides (e.g., 15 bases to 40 bases) are readily performed in an automated fashion via oligonucleotide synthesizers using commercially available columns, or reactors, containing solid supports specific for oligonucleotide synthesis, such as controlled pore glass (CPG) solid supports. However, there is a growing demand for large-scale synthesis of longer oligonucleotides, such as oligonucleotides for CRISPR protocols that are routinely over 80 bases long. Current synthesizers and columns are not optimized for large scale/long oligonucleotide preparations, which can lead to several problems during synthesis. For example, because step-wise efficiency in each cycle of synthesis is typically less than 100% the longer the oligonucleotide, the lower the overall efficiency, and the number of undesired oligomers (e.g., truncations, deletions, and depurinations) increases. The synthesizer and columns may not

appropriately handle the larger quantities of materials, resulting in fouling or clogging of the fluid distributors, diffusers, and the solid support, which can also lead to high back pressures. High back pressures can lead to part failures, changes in reaction rates, incomplete synthesis, and under or over exposure of the solid support to reagents. Further, columns in large scale synthesis are susceptible to nonuniform packing of the solid support and/or resins inside the column, causing void formation and/or flow channeling. These columns are also prone to form nonuniform distribution of fluids and reagents throughout the solid support, leading to the solid supports and/or resins being exposed to reagents at different times and at different concentrations, resulting in reduced yields. Therefore, a device, system, or method of increasing the efficiency of large scale synthesis of long oligonucleotides over current devices, systems and methods is desired.

SUMMARY

(3) A reactor system is disclosed. The reactor system includes a vessel configured to contain a solid support, the vessel including: a vessel wall defining a reaction chamber, the reaction chamber having a first end and a second end opposite the first end; a piston operatively arranged at the first end and configured to translate within the reaction chamber; a force measuring device coupled to the piston and configured to measure a load on the piston; a piston driver coupled to the piston; and a processor operably coupled to the force measuring device and the piston driver, the processor configured to: measure a load on the piston using the force measuring device, and adjust a position of the piston using the piston driver based on measuring the load on the piston. In various embodiments of the reactor system, the piston may include a piston head and a piston shaft, where the force measuring device may be coupled to the piston shaft.

(4) Various embodiments of the reactor system may include an inlet disposed at the first end of the reaction chamber and configured to receive an influent; and/or an outlet disposed at the second end of the reaction chamber and configured to release an effluent. Some embodiments of the reactor system may further include a flow distribution unit coupled to the piston and disposed within the vessel adjacent the reaction chamber, wherein the flow distribution unit includes a plurality of flow distribution channels, wherein a first end of each of the plurality of flow distribution channels is fluidly coupled to the inlet, and wherein a second end of each of the plurality of flow distribution channels is fluidly coupled to the reaction chamber. In particular embodiments of the reactor system, the flow distribution unit may include a modular flow distribution unit including a distributor head coupled to a distributor cap with a filter disposed therebetween, wherein the plurality of flow distribution channels is disposed within the distributor head. In some embodiments of the reactor system the flow distribution unit may be an inlet flow distribution unit and the reactor system may further include an outlet flow distribution unit which includes a modular flow distribution unit coupled to the outlet.

(5) Some embodiments of the reactor system may further include a pressure sensor operatively coupled to the processor and configured to measure a pressure within the vessel, and the processor may be further configured to obtain a measurement of the pressure within the vessel from the pressure sensor, and adjust the pressure within the vessel based on the measurement of the pressure including at least one of adjusting the position of the piston or actuating a pressure-control device including a valve.

(6) Various embodiments of the reactor system may further include a solid support disposed within the reaction vessel, and a distance sensor operatively coupled to the processor and configured to measure a distance between the piston and the solid support, and the processor may be further configured to obtain a measurement of the distance between the piston and the solid support, and adjust the position of the piston based on the measurement of the distance. In some embodiments of the reactor system, information regarding the distance between the piston and the solid support may be obtained using information from the force measuring device coupled to the piston, which can be determined at least in part on whether the piston is or has been in contact with the solid support. In certain embodiments of the reactor system the solid support may be configured to support at least

one of: oligonucleotide synthesis, pretreatment, cleavage, deprotection, or purification.
(7) In certain embodiments of the reactor system, the piston may be a first piston and the reactor system may further include a second piston operatively arranged at the second end of the reaction chamber and configured to translate within the reaction chamber. In particular embodiments of the reactor system, the vessel may further include at least one of: a sampling port, a loading port configured to facilitate loading of the solid support into the reaction chamber, or an unloading port configured to facilitate unloading of the solid support out of the reaction chamber.
(8) In various embodiments of the reactor system, the force measuring device may include at least one of a load cell or a strain gauge and the piston driver may drive the piston using at least one of mechanical, hydrodynamic, or pneumatic force. In certain embodiments of the reactor system, the load on the piston measured by the force measuring device may be proportional to a pressure within the reaction chamber.
(9) Particular embodiments of the reactor system may further include a sensor coupled to the inlet, wherein the sensor may include at least one of a flow sensor, a pH sensor, a conductivity sensor, a thermal sensor, a Raman sensor, a mid-IR sensor, or a spectrometer including UV spectrometer. Some embodiments of the reactor system may further include a fluid conditioner fluidly coupled to the inlet, wherein the fluid conditioner may include at least one of a heat exchanger, a degasser, a de-bubbler, or a gas dryer.
(10) In some embodiments of the reactor system, the piston may include a flow distribution plate including a plurality of pores. In various embodiments of the reactor system, the plurality of pores of the flow distribution plate may include different size openings, and the different size openings of the plurality of pores may be distributed in a radial pattern across the flow distribution plate.
(11) In particular embodiments of the reactor system, the processor may be further configured to: collect a plurality of real-time data based on obtaining a plurality of measurements of the load on the piston using the force measuring device, wherein the plurality of real-time data may indicate at least one of a reaction chemistry or flow dynamics within the reaction chamber, and adjust the position of the piston using the piston driver based on collecting the plurality of real-time data to adjust at least one of a system or operational parameter in real time.
(12) In some embodiments of the reactor system, the vessel may further include a piston head disposed at an end of the piston and a plurality of crossflow ports coupled to the piston head, where the plurality of crossflow ports may be configured to add or remove fluid from a space between the piston head and a solid support disposed within the reaction chamber.
(13) In certain embodiments of the reactor system, the piston head may include an inner piston head and a porous outer piston head that is configured to be disposed adjacent a solid support disposed within the reaction chamber. In some embodiments of the reactor system, the plurality of crossflow ports may be configured to circulate fluid in a space between the inner piston head and the porous outer piston head to reduce or prevent fouling or clogging of the porous outer piston head. In particular embodiments of the reactor system, the porous outer piston head may be or include a filter screen. In various embodiments of the reactor system, the inner and outer piston heads may be adjustable relative to one another or may be fixed with the result that the space between the inner and outer pistons, sometimes referred to herein as head space, may be adjustable or may be fixed.
(14) Some embodiments of the reactor system may further include an inlet disposed at the first end of the reaction chamber and configured to receive an influent. The vessel may further include an inner piston head disposed at an end of the piston, where the inner piston head may include a perforated inlet distributor, and a porous outer piston head configured to be disposed adjacent a solid support disposed within the reaction chamber. The inlet may be coupled to the perforated inlet distributor and the perforated inlet distributor may be configured to direct fluid from the inlet in a plurality of directions into a space between the inner piston head and the porous outer piston head.
(15) In various embodiments of the reactor system, the vessel may further include an inner piston

head disposed at an end of the piston and a porous outer piston head configured to be disposed adjacent a solid support disposed within the reaction chamber. The space between the inner piston head and the porous outer piston head may include a head space and a volume of the head space may be fixed or adjustable. In certain embodiments of the reactor system, the head space may include at least one of a filter, a porous material, a screen, or a reagent. In some embodiments which include fixed head space, the vessel may not include a porous outer piston. Instead, in certain embodiments the porous outer piston may be replaced by a filter screen and the fixed head space may be directly designed into the inner piston head itself. Therefore, in particular embodiments of the system, the vessel may further include an inner piston head disposed at an end of the piston and a filter screen configured to be disposed adjacent a solid support disposed within the reaction chamber, where the space between the inner piston head and the filter screen may include a head space and where a volume of the head space may be fixed.
(16) In various embodiments of the reactor system, an outer wall of the vessel may be in contact with a heat exchanger which at least one of heats or cools the vessel. The heat exchanger may include at least one of fluid circulation, microwave heating, inductive heating, or Peltier heating/cooling. In certain embodiments of reactor system, an outer wall of the vessel may be in contact with an ultrasonic generator which is configured to agitate the vessel.
(17) A method of operating a reactor is disclosed. The method may include providing a vessel configured to contain a solid support, wherein the vessel may include a vessel wall defining a reaction chamber, the reaction chamber having a first end and a second end opposite the first end, a piston operatively arranged at the first end and configured to translate within the reaction chamber, a force measuring device coupled to the piston and configured to measure a load on the piston, and a piston driver coupled to the piston, and a processor operably coupled to the force measuring device and the piston driver; measuring, using the processor, a load on the piston using the force measuring device; and adjusting, using the processor, a position of the piston using the piston driver based on measuring the load on the piston.
(18) In some embodiments of the method, providing a vessel may further include providing an inlet disposed at the first end of the reaction chamber and configured to receive an influent, and providing an outlet disposed at the second end of the reaction chamber and configured to release an effluent; and the method may further include at least one of: delivering the influent to the reaction chamber via the inlet, or obtaining the effluent from the reaction chamber via the outlet.
(19) In various embodiments of the method providing the vessel may further include providing the force measuring device including at least one of a load cell or a strain gauge, and providing the piston driver wherein the piston driver drives the piston using at least one of mechanical, hydrodynamic, or pneumatic force.
(20) In particular embodiments of the method, providing the vessel may further include providing a pressure sensor coupled to the reaction chamber and the method may further include measuring a pressure within the reaction chamber using the pressure sensor. In some embodiments of the method, upon measuring an increase in pressure, adjusting the position of the piston may further include adjusting the position of the piston to increase a volume of the reaction chamber and reduce a pressure within the reaction chamber. In some embodiments of the method, upon measuring a decrease in pressure, adjusting the position of the piston may further include adjusting the position of the piston to decrease a volume of the reaction chamber and increase a pressure within the reaction chamber.
(21) In various embodiments of the method, providing the vessel may further include providing a sensor coupled to the inlet, wherein the sensor may include at least one of a flow sensor, a pH sensor, a conductivity sensor, a thermal sensor, a Raman sensor, a mid-IR sensor, or a spectrometer; and delivering the influent to the reaction chamber may further include measuring a property of the influent using the sensor.
(22) In some embodiments of the method, providing the vessel may further include providing a

fluid conditioner coupled to the inlet, wherein the fluid conditioner may include at least one of a heat exchanger, a degasser, a de-bubbler, or a gas dryer; and wherein delivering the influent to the reaction chamber may further include adjusting a property of the influent using the fluid conditioner.

(23) In certain embodiments of the method, providing the vessel may further include providing a flow distribution unit coupled to the piston and disposed within the vessel adjacent the reaction chamber, wherein the flow distribution unit is fluidly coupled to the inlet; and wherein delivering the influent to the reaction chamber may further include delivering the influent to the reaction chamber using the flow distribution unit. In various embodiments of the method, providing the flow distribution unit may further include providing the flow distribution unit, wherein the flow distribution unit may include a plurality of flow distribution channels, wherein a first end of each of the plurality of flow distribution channels is fluidly coupled to the inlet, and wherein a second end of each of the plurality of flow distribution channels is fluidly coupled to the reaction chamber; and wherein delivering the influent to the reaction chamber may further include delivering the influent to the reaction chamber using each of the plurality of flow distribution channels of the flow distribution unit.

(24) A modular flow distribution unit is provided. The modular flow distribution unit may include a distributor head include a central channel at a first end; a distributor cap coupled to a second end of the distributor head opposite the first end; and a filter disposed between the distributor head and the distributor cap at the second end of the distributor head. In some embodiments of the modular flow distribution unit, the distributor head may include a plurality of flow distribution channels, wherein a first end of each of the plurality of flow distribution channels is fluidly coupled to the central channel at the first end of the distributor head, and wherein a second end of each of the plurality of flow distribution channels is fluidly coupled to the second end of the distributor head. In some embodiments of the modular flow distribution unit, the plurality of flow distribution channels may branch outward from a central outlet. In various embodiments of the modular flow distribution unit, the distributor head and the distributor cap may be made from chemical-resistant materials at least one of natural PEEK, stainless steel, or Teflon. In some embodiments of the modular flow distribution unit, the distributor head may further include at least one input port to provide for at least one of a cross flow or a sensor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description is described with reference to the accompanying figures. The use of the same reference numbers in different instances in the description and the figures may indicate similar or identical items. Various embodiments or examples of the present disclosure are disclosed in the following detailed description and the accompanying drawings. The drawings are not necessarily to scale. In general, operations of disclosed processes may be performed in an arbitrary order, unless otherwise provided in the claims.

(2) FIG. **1** is a flow diagram of an oligomer synthesis cycle, in accordance with one or more embodiments of the disclosure.

(3) FIG. **2**A is a drawing illustrating a perspective view of a reactor, in accordance with one or more embodiments of the disclosure.

(4) FIG. **2**B is a drawing illustrating a cutaway side view of the reactor of FIG. **2**A, in accordance with one or more embodiments of the disclosure.

(5) FIG. **3** provides a diagram of a reactor system in accordance with one or more embodiments of the disclosure.

(6) FIG. **4** is a block diagram illustrating a reactor system, in accordance with one or more

embodiments of the disclosure.
(7) FIGS. **5**A-**5**B are drawings illustrating cutaway-side views of a reactor, in accordance with one or more embodiments of the disclosure.
(8) FIG. **6**A is a drawing illustrating cutaway-side views of a reactor, in accordance with one or more embodiments of the disclosure.
(9) FIGS. **6**B-**6**F are drawings illustrating cutaway-side views of a reactor, in accordance with one or more embodiments of the disclosure.
(10) FIGS. **7**A-**7**D show drawings illustrating use of a flow distributor with a reactor, in accordance with one or more embodiments of the disclosure.
(11) FIG. **8** is a drawing illustrating a cutaway side view of a reactor operatively coupled to a diffuser housing and a degasser housing, in accordance with one or more embodiments of the disclosure.
(12) FIGS. **9**A-**9**D are drawings illustrating cutaway side views of a reactor, in accordance with one or more embodiments of the disclosure.
(13) FIGS. **10**A-**10**B are drawings illustrating cutaway-side views of a reactor, in accordance with one or more embodiments of the disclosure.
(14) FIG. **11**A is a drawing illustrating a cutaway side view of a reactor operatively coupled to a signal generator, in accordance with one or more embodiments of the disclosure.
(15) FIG. **11**B is a drawing illustrating a plan view of a reactor, in accordance with one or more embodiments of the disclosure.
(16) FIG. **12** is a drawing illustrating a cutaway side view of a reactor, in accordance with one or more embodiments of the disclosure.
(17) FIG. **13** is a drawing illustrating a cutaway side view of a reactor, in accordance with one or more embodiments of the disclosure.
(18) FIGS. **14**A-**14**B are drawings illustrating cutaway side views of a reactor, in accordance with one or more embodiments of the disclosure.
(19) FIG. **15**A is a drawing illustrating a cutaway side view of a reactor, in accordance with one or more embodiments of the disclosure.
(20) FIGS. **15**B-**15**D are drawings illustrating cutaway side views of a reactor operatively coupled to a heat exchanger, in accordance with one or more embodiments of the disclosure.
(21) FIGS. **16**A-**16**D are drawings illustrating a degasser module, in accordance with one or more embodiments of the disclosure.
(22) FIG. **17** is a drawing illustrating a cutaway side view of a reactor, in accordance with one or more embodiments of the disclosure.
(23) FIGS. **18**A-**18**B are drawings illustrating cutaway side views of a reactor and a resin-filled scaffold/frame placed within the reactor, in accordance with one or more embodiments of the disclosure.
(24) FIGS. **19**A-**19**B are a perspective and partially transparent views of a reactor, in accordance with one or more embodiments of the disclosure.
(25) FIGS. **20-22** are flow schemes for polymer synthesis that include at least one reactor, in accordance with one or more embodiments of the disclosure.
(26) FIG. **23** is a flowchart illustrating a method for transferring an influent through a reactor, in accordance with one or more embodiments of the disclosure.
(27) FIGS. **24-29** show constructions of a modular flow distributor in accordance with one or more embodiments of the disclosure.
(28) FIG. **30** shows data collected from a synthesis system at different stages during oligonucleotide synthesis.
(29) FIG. **31** shows data collected from a synthesis system at different stages during oligonucleotide synthesis, comparing a static column (left) to a column that is dynamically adjusted during operation (right).

(30) FIG. **32** shows a position of a piston during operation of a dynamic column.
(31) FIGS. **33**A-**33**D show data indicating quality and quantity of a 100-bp oligo synthesized using several known systems (labeled "control") compared to the oligo prepared using the presently-disclosed procedures (labeled "current invention").
(32) FIGS. **34**A-**34**B show HPLC data for oligos prepared using dynamic (FIG. **34**A) or static (FIG. **34**B) pistons.

DETAILED DESCRIPTION

(33) Before the present disclosure is described in greater detail, it is to be understood that this disclosure is not limited to particular embodiments described, and as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the present disclosure will be limited only by the appended claims.
(34) As used herein a letter following a reference numeral is intended to reference an embodiment of the feature or element that may be similar, but not necessarily identical, to a previously described element or feature bearing the same reference numeral (e.g., 1, 1a, 1b). Such shorthand notations are used for purposes of convenience only and should not be construed to limit the disclosure in any way unless expressly stated to the contrary.
(35) Further, unless expressly stated to the contrary, "or" refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by anyone of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).
(36) In addition, use of "a" or "an" may be employed to describe elements and components of embodiments disclosed herein. This is done merely for convenience and "a" and "an" are intended to include "one" or "at least one," and the singular also includes the plural unless it is obvious that it is meant otherwise.
(37) Finally, as used herein any reference to "an embodiment", "one embodiment" or "some embodiments" means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment disclosed herein. The appearances of the phrase "in some embodiments" in various places in the specification are not necessarily all referring to the same embodiment, and embodiments may include one or more of the features expressly described or inherently present herein, or any combination of sub-combination of two or more such features, along with any other features which may not necessarily be expressly described or inherently present in the instant disclosure.
(38) Described herein are embodiments of a reactor, or column, for the synthesis of oligonucleotides, and of oligonucleotide synthesizer systems configured to use the reactor. The reactor includes a vessel configured to hold a solid support, such as a solid support for oligonucleotide synthesis. The reactor includes an inlet for receiving an influent, and an outlet for releasing an effluent. The reactor further includes a first end, and a first piston operatively arranged at the first end, configured to translate within the vessel. The first piston is configured such that the adjustment of the first piston within the vessel may improve the packing density and flow characteristics of the influent through the solid support. The reactor may also include a second end, and a second piston operatively arranged at the second end, configured to translate within the vessel. Similar to the first piston, the adjustment of the second piston may also improve the packing density and flow characteristics of the influent through the solid support. Systems and methods that include the reactor are also described.
(39) A reactor (and the supports therein, such as solid supports) according to various embodiments may be used for the purification and/or synthesis and post-processing of products and may be used for the isolation of single chemical compounds from a mixture, such as isolation of a single oligonucleotide that is synthesized within the reactor or the isolation of plasmid DNA from a bacterial preparation, as well as steps of pretreatment, cleavage, deprotection, and/or purification.

In this regard, the reactor may be similar to a column, such as a column used in column chromatography. In an embodiment, the reactor may be configured for oligonucleotide synthesis. For instance, the reactor may be couplable to an oligonucleotide synthesis system, where the reactor is the site of oligonucleotide synthesis. In another example, the reactor may be configured for peptide synthesis. In another example, the reactor may be used for purification of proteins or nucleic acids, such as the purification of affinity tagged proteins or plasmids from a solution containing cellular material (e.g., lysed cells). The reactor includes a vessel that further includes a vessel wall surrounding a hollow space (e.g., the vessel wall forming a hollow three dimensional shape, such as a hollow cylinder). The vessel houses the solid support or other matrix required for oligonucleotide synthesis or other synthesis/purification. The vessel may have a size, shape, or dimension of any type of open container, such as the shape of a hollow cylinder as shown in FIGS. **2**A and **2**B. Embodiments of the disclosed device may be made in a compact and/or enclosed housing that may be suitable for placement in a laboratory, e.g., on a benchtop.

(40) The vessel includes a first end disposed at one end of the shape formed by the vessel wall (e.g., one end of the hollow cylinder). The reactor may further include a first piston operatively arranged at the first end. The first piston is configured to create a seal along the inner circumference of the vessel and translate back and forth along the length of the vessel. The reactor is configured such that first piston may be translated to a position relative to the solid support that may improve a flow characteristic of the reactor. Improved flow characteristics include but are not limited to lower influent pressure (e.g., lower back pressure), improved uniform flow across the solid support, increased flow, decreased void formation, and decreased head space (e.g., decreased space between the piston head of the first piston and the solid support). The first piston may also be used to push the influent through the solid support.

(41) In some embodiments, the vessel further includes a second end disposed opposite of the first end. The reactor may include a second piston operatively arranged at the second end. Similar to the first piston, the second piston may be configured to create a seal along the inner circumference of the vessel and translate back and forth along the length of the vessel and may be translated to a position relative to the solid support that may improve flow characteristics of the reactor as mentioned above. The second piston may also be used to push and/or pull the influent and through the solid support.

(42) The reactor further includes an inlet configured to receive an influent, and an outlet configured to release an effluent. However, the inlet may at times release an effluent, and the outlet may at time receive an influent, depending on the direction that fluid is being pumped through the reactor. The inlet and the outlet are operatively coupled to the first piston and second piston such that the fluid travels through a port located on or adjacent to the pistons. The inlet and/or the outlet may be disposed at a stem of the piston or may be disposed along a non-stem portion.

(43) In embodiments, the reactor further includes a loading port configured to facilitate loading of solid support into the vessel, and an unloading port configured to facilitate unloading of the solid support out of the vessel. For example, after synthesis and harvest of an oligonucleotide from the reactor, the solid support may be removed from the vessel via the unloading port, and fresh solid support may be inserted into the vessel via the loading port. The loading and unloading of the solid support may be performed manually. For example, the solid support may be injected and removed as a slurry using a hand-held syringe. In another example, the solid support may be transferred in and out of the vessel using pumps. The solid support may be of any type of material or matrix capable of supporting oligonucleotide synthesis including but not limited to controlled pore glass (CPG) and polystyrene. In other embodiments, the reactor further includes in-process sampling ports (FIG. **17**). That is, samples of the solid support may be collected manually or automatically from the column at any step of synthesis. In other embodiments, the reactor further includes washing ports. These ports may be used to manually or automatically wash the reactor before and/or after each use.

(44) It should be noted that reactor may be configured to contain any type of stationary phase and is not limited to containing a solid support. The reactor may be configured for any type of column or column bed technology including but not limited to solid phase, liquid phase, batch, continuous flow, packed bed, and fluidized bed technologies. Although the term "solid support" is used within this document, it should be understood that the term "solid support" may be used interchangeably with any stationary phase element from any technology, such as the technologies listed above.
(45) FIGS. **2**A-**2**B are diagrams illustrating a reactor **300**, in accordance with one or more embodiments of the disclosure. The reactor **300** may include one or more, or all, components of reactor **300**, and vice versa. The reactor **300** includes a top inner piston **304**, and a top outer piston **308**, and may further include a bottom inner piston **312** and a bottom outer piston **316**. One or more of the outer pistons **308**, **316** act to adjustably contain the solid support, while one or more inner pistons **304**, **312** adjust the volume of a head space **320** between a porous outer piston head **324** (also referred to as a distribution plate **325**) of the outer piston **308**, **316** and an inner piston head **326** of the inner piston **304**, **312**. The inner piston **304**, **312** may be configured such that an inner piston stem **328** of the inner pistons **304**, **312** fits within the outer stem **332** of the outer piston **308**, **316**, allowing the inner piston stem **328** to translate independently of the outer piston stem **332**. The inner piston stem **328** and outer piston stem may be operatively coupled (e.g., via a screw thread mechanism or other type of mechanism) so that the head space **320** may be adjusted by manual or automatic means. The inlet **236** and outlet **237** may be disposed within the inner piston stem **328** of the top inner piston **304** and bottom inner piston **312**, respectively.
(46) In embodiments, the reactor **300** includes two or more crossflow ports **336***a-d* configured to move fluid in and out between the inner piston head **326** and the outer piston head **324** or solid support. Fluid entering and exiting the crossflow ports **336***a-d* wash and/or rinse a volume inside the vessel **204**, preventing or actively treating clogging and/or fouling within the reactor **300**. The reactor **300** may also include one or more vents **340***a-b* configured to allow the transfer of gas into, or out of, the reactor **300**. For example, an inert gas may be introduced into the reactor **300** via a vent **340***a* (or via inlet **236** and/or outlet **237**) in an effort to uniformly push liquid/reagent through the solid support. Gas driven flow, also known as pressure-driven flow, can result in improved and consistent flow dynamics within the reactor **300**, as shown in FIGS. **5**A-**5**B. For example, in conventional volumetric flow-rate driven flow with a single influent port at the center, influent is often unable to fully distribute itself along the length of the distribution plate **325**, as shown in FIG. **5**A, with a higher concentration of reagent flowing through the middle of the reactor **300**, as indicated by the arrows **348**. In pressure-driven flow, the influent enters into an open space within or adjacent to a first piston **220** (FIG. **6**) and is able to equilibrate within the available space at the top of the reactor **300**. After equilibration, gas (e.g., nitrogen gas) is directed **325** into the reactor **300**, and the pressure of the gas forces the influent through the reactor **300** in an equilibrated and constant fashion **348** (FIG. **5**B). As shown in FIG. **5**B, the addition of a modular flow distribution unit (see further discussion below) on the top and/or bottom further helps to evenly distribute fluid in the column.
(47) FIG. **3** provides a diagram of a reactor system **3300** in accordance with one or more embodiments of the disclosure. The system **3300** includes a piston driver **1** which may be a mechanically, hydrodynamically, and/or pneumatically-driven system to drive the piston up and/or down. The piston driver **1** is coupled to and driven by a linear shaft **2**. The system **3300** may include one or more in-line sensor **3** which may be in-line with the input and/or output fluid lines and which may include one or more of a flow sensor, a pH sensor, a conductivity sensor, a thermal sensor, a Raman sensor, and/or a mid-IR sensor. The system **3300** may also include one or more in-line fluid conditioner **4** which may be in-line with the input and/or output fluid lines and which may include one or more of a heat exchanger, a degasser, a de-bubbler, and/or a gas dryer.
(48) In various embodiments, the system **3300** may further include a force measuring device **5** in series with the piston **2**, where the force measuring device **5** may include one or more of a load cell

and/or a strain gauge. The system **3300** may also include one or more chamber sensor **6** coupled to the top and/or bottom of the chamber (e.g., via connections embedded in the piston(s) or upper/lower wall(s) of the chamber) which may include one or more of a differential pressure sensor, a pressure sensor, a strain gauge, a pH sensor, a conductivity sensor, a thermal sensor, an optical sensor, a Raman sensor, and/or a mid-IR sensor.

(49) In certain embodiments, the system **3300** may include a control system **7** which monitors the various components of the system **3300**, e.g., in a closed-loop manner. The control system **7** may be coupled (e.g., in a wired and/or wireless manner) to one or more of piston driver **1**, in-line sensor **3**, fluid conditioner **4**, force measuring device **5**, chamber sensor **6**, pressure relief/control valve **8**, and/or leak detector/limit switch sensor **14** (FIG. **3**). In some embodiments, the system **3300** may include one or more valve **8** such as a pressure relief valve and/or control valve, for example connected in-line to the input and/or output fluid lines.

(50) In some embodiments, the control system **7** (including, e.g., a processor) may be operably coupled to various components such as the force measuring device **5** and/or the piston driver **1** (or other components) and may be configured to measure a load on the piston **2** using the force measuring device **5**, and adjust a position of the piston **2** using the piston driver **1** based on measuring the load on the piston **2**. In other embodiments, the control system **7** may be configured to collect a plurality of real-time data based on obtaining a plurality of measurements (e.g., measurements of the load on the piston **2** or any other measurements using various devices), with the result that the plurality of real-time data is indicative of conditions within the reaction chamber during a reaction, for example indicative of at least one of a reaction chemistry or flow dynamics within the reaction chamber. The control system **7** may then be configured to adjust the position of the piston **2** using the piston driver **1** based on collecting the plurality of real-time data to adjust at least one of a system or an operational parameter in real time.

(51) In various embodiments, the system **3300** may include a synthesis column including a reaction chamber defined on the bottom by a modular bottom flow distribution unit **9** and on the top by a modular top flow distribution unit **10**, where either or both of the units **9**, **10** may be reusable. Each unit **9**, **10** may include one or more flow distribution channels **9***a*, **10***a* to evenly distribute and/or collect the fluid moving through the reaction chamber. A first end of each of the flow distribution channels **9***a*, **10***a* is fluidly coupled to a fluid inlet or outlet and a second end of each of the flow distribution channels **9***a*, **10***a* is fluidly coupled to the reaction chamber. The synthesis column includes an outer housing **11** which may be made of rigid and non-reactive materials such as glass and/or stainless steel. The flow distribution units **9**, **10** form a tight fit within the outer housing **11** in order to minimize or prevent leakage of reaction materials from the reaction chamber.

(52) The reaction chamber may include one or more leak detector/limit switch sensor **14** associated with and/or adjacent to the units **9**, **10** to sense fluid leaks and/or to detect upper or lower limits of movement of the units **9**, **10**. The reaction chamber may include a solid state support **12** disposed therein, where the support **12** may include controlled pore glass (CPG) and/or polymeric resins. In particular embodiments, the system **3300** is configured to evenly distribute fluid throughout the reaction chamber to create a uniform flow distribution **13** across the column diameter.

(53) In various embodiments, the system **3300** is configured to provide various improvements over known systems. In some embodiments, the system **3300** is configured to provide automated initial packing and to continuously maintain the resin/solid support packing density throughout the entire synthesis using in-line load cells. In other embodiments, the system **3300** is configured to provide uniform flow distribution across the entire diameter of the column, e.g., using the disclosed innovative flow distribution units along with cleanable and reusable filters. In still other embodiments, the system **3300** is configured to continuously monitor and dynamically respond to the real-time conditions inside the column.

(54) In particular embodiments, the primary source of information about the condition inside the column may come from the force measuring device **5** (e.g., a load cell), which provides

information about force exerted on the piston **2**, which may also be confirmed by measurements of differential pressure across the resin/solid support (obtained for example using chamber sensor **6**). In some embodiments, the system **3300** may be configured to dynamically adjust the height of the column (e.g., by using the piston driver **1**) to accommodate both swellable and non-swellable resin/solid supports. Making these dynamic adjustments improves performance at least by avoiding backpressure, which is a common problem in synthesis columns (particularly when synthesizing longer oligonucleotides), for example by using pressure relief/control valve **8** and/or by dynamically adjusting the height of the column, which may be controlled by control system **7**. Further improvements in performance may also be realized using efficient and on-demand heating of reagents by using the in-line fluid conditioner **4** which may include a heat exchanger, which helps to avoid formation of high temperature gradients across the diameter of the column.

(55) In various embodiments of the system **3300**, the column has a symmetrical design meaning that the influent and effluent to and from the column can be swapped as well as instruments before and after the column. In certain embodiments, either one or both of the top and bottom flow distribution units **9**, **10** can move dynamically, where the flow distribution units **9**, **10** provide uniform flow distribution across the whole diameter of column.

(56) Although in various embodiments disclosed herein the reactor is shown as being in a fixed position while one or both pistons are shown as being movable, in some embodiments a piston may be held in a fixed position and the reactor may be moved relative to the piston to make adjustments to the reaction volume.

(57) FIG. **4** is a block diagram illustrating a reactor system **242** in accordance with one or more embodiments of the disclosure. The reactor system **242** may be used for the synthesis and/or purification of oligonucleotides and/or peptides/proteins. For example, the reactor system **242** may be configured as an oligonucleotide synthesis system. The reactor system **242** includes one or more sensors **244***a-b* operatively coupled to a reactor vessel **204** and configured to sense one or more characteristics of the reactor vessel **204** or influent within the reactor vessel **204**. For example, the sensors **244***a-b* may be configured to sense or detect an influent pressure, an effluent pressure, pressure within the reactor vessel **204**, piston force (e.g., biasing force), conductance, pH, distance between the piston **220**, **232** and the solid support, absorbance and/or temperature. The sensors **244***a-b* may also be configured to sense or detect characteristics of the fluid before the fluid enters the reactor vessel **204** or after the fluid leaves the reactor vessel **204** including but not limited to pressure, temperature, and absorbance. The sensors **244***a-b* may be disposed within or adjacent to the reactor vessel **204**. The reactor system **242** further includes one or more regulators **248** operatively coupled to the reactor vessel **204** and configured to regulate influent entering the reactor vessel **204**. For example, the one or more regulators **248** may include pumps or pump controllers to regulate fluid flow. In another example, the one or more regulators **248** may include one or more valves configured to switch the influent entering the reactor vessel **204**.

(58) In embodiments, the reactor system **242** includes a piston driving circuit **252** operatively coupled to the reactor vessel **204** and configured to control the movement of the one or more pistons **220**, **232**. The piston driving circuit **252** may be operatively coupled to the one or more sensors **244***a-b* and/or regulators **248** such that the piston driving circuit **252** operates based on information (e.g., feedback) received from the sensors **244***a-b*. For example, the reactor system **242** may further include a controller **256** operatively coupled to the piston driving circuit **252** that provides processing functionality to the sensors **244***a-b*, the regulators **248**, the piston driving circuit **252**, and/or other components of the reactor system **242**. The controller **256** includes processors **260**, memory **264**, and a communication interface **268**.

(59) The controller **256** may include one or more processors **260** (e.g., micro-controllers, circuitry, field programmable gate array (FPGA), programmable logic controllers (PLCs), modular fieldbus, central processing units (CPU), application-specific integrated circuit (ASIC), phidgets, or other processing systems), and resident or external memory **264** for storing data, executable code, and

other information. The controller **256**, via the one or more processors **260**, may execute one or more software programs embodied in a non-transitory computer readable medium (e.g., memory **264**) that implement techniques described herein. The controller **256** is not limited by the materials from which it is formed or by the processing mechanisms employed therein and, as such, can be implemented via semiconductor(s) and/or transistors (e.g., using electronic integrated circuit (IC) components), and so forth.

(60) The memory **264** can be an example of tangible, computer-readable storage medium that provides storage functionality to store various data and/or program code associated with operation of the controller **256**, such as software programs and/or code segments, or other data to instruct the controller **256**, and possibly other components operatively coupled to the controller **256** to perform the functionality described herein. Thus, the memory **264** can store data, such as a program of instructions, and so forth (e.g., the instructions causing the one or more processors to implement the techniques described herein). It should be noted that while a single memory **264** is described, a wide variety of types and combinations of memory **264** (e.g., tangible, non-transitory memory) can be employed. The memory **264** may be integral within the controller **256**, may include stand-alone memory, or may be a combination of both. Some examples of the memory **264** may include removable and non-removable memory components, such as random-access memory (RAM), read-only memory (ROM), flash memory (e.g., a secure digital (SD) memory card, a mini-SD memory card, and/or a micro-SD memory card), solid-state drive (SSD) memory, magnetic memory, optical memory, universal serial bus (USB) memory devices, hard disk memory, external memory, and so forth.

(61) The communication interface **268** may be operatively configured to communicate with components of the controller **256**. For example, the communication interface **268** can be configured to retrieve data from the controller **256** or other components, transmit data for storage in the memory **264**, retrieve data from storage in the memory **264**, and so forth. The communication interface **268** can also be communicatively coupled with the controller **256** to facilitate data transfer between components of the controller **256**. The controller **256** may also include and/or connect to one or more input/output (I/O) devices (e.g., a graphical user interface or GUI). In embodiments, the communication interface **268** includes or is coupled to a transmitter, receiver, transceiver, physical connection interface, or any combination thereof.

(62) FIG. **6**A is a drawing illustrating various stages of operation of a reactor **420**, where each stage of operation is facilitated by setting a specific load setpoint and bandwidth (e.g., measured by a force measuring device **225** such as a load cell associated with the piston **220**) for that particular step or stage. The stages include one or more of: a loading/fill stage **400**, a touch stage **402**, a compress/packing stage **404**, a flow stage **408**, a decompress stage **412**, and/or an unloading stage **414** (homing) of the reactor **420**, in accordance with one or more embodiments of the disclosure. Reactor **420** is shown having a top piston **220** which includes a piston shaft **221**, a built-in inlet **236**, a force measuring device **225** (e.g., a load cell) coupled to the piston shaft **221**, and a porous piston head **224**, which acts as a diffuser (e.g., a distributor plate or frit), although in various embodiments reactor **420** may also include a second/lower piston. At fill stage **400**, a solid support **424**, such as resin, is added to the reactor **420** (e.g., via a loading port and an unloading port) so that the loaded solid support **424** reaches a fill level **428**; at this stage, the load cell reading is very low, e.g., almost 0 N or otherwise a value much lower than that of set for other stages such as the touching stage. In general, the force measuring device may have a range of at least 0 to 20 kN or more with each target force level at various stages being in a range of +/−10% of the target value.

(63) Once filled, the reactor **420** advances to the touch stage **402**. At the touch stage **402**, the piston is commanded to move downward until the force measuring device **225**/load cell reaches a force value set in the program for the touch stage **402**, e.g., a force maximum for completion of the touch stage **402** (e.g., 300 N). Next, the reactor **420** may progress to the compress stage **404**, wherein one or more pistons **220** press upon the solid support **424**, where the progress and successful

completion of each stage may be monitored and determined by comparison of the output of the force measuring device **225** with the specific load setpoint and bandwidth. At the compress/packing stage **404**, the piston **220** is commanded to further move down into the chamber until the force measuring device **225**/load cell reaches a force value set in the program for the compress/packing stage (e.g., 500 N). This extra compression (beyond the force applied at the touch stage **402**) helps to uniformly pack the solid support **424** and minimize any void formation during the long flow stage **408**. Thus, compressing the solid support **424** provides benefits to the reactor **420** including but not limited to reducing voids/channels in the solid support **424** and reducing head space between the piston head **224** and the solid support **424**. Compression may also increase the packing density of the solid support **424**, which may increase the efficiency with which the solid support **424** binds/interacts with flow components.
(64) After compression, the reactor **420** may advance to flow stage **408**, where influent is added to the reactor **420**. The main synthesis stage, which includes several cycles and several steps in each cycle, starts at the flow stage **408**. At the flow stage **408**, the load on the piston **220** is continuously measured and the load values are communicated to the same control system that is used to control the piston actuator. As a result, the piston **220** is dynamically adjusted to maintain packing of the solid support **424** throughout the synthesis process. That is, the load remains within a defined range that is set at the compress/packing stage **404** (e.g., 500 N).
(65) After the flow stage **408**, the reactor **300** may enter a decompress stage **412**, where the one or more pistons **220** are moved so as to increase the volume of the reactor **420** that contains the solid support **424**. Decompression also may reduce back pressure caused by the reduction in voids or flow path between the support particles as, for example, the length of oligomer increases.
(66) During the entire course of synthesis, the load on the piston **220** varies at each step of a cycle and also between cycles. This load variation can be used not only to understand the real time synthesis chemistry and conditions inside the column/reactor, but it can also be used as feedback to logically adjust the piston's position (e.g., at the decompress stage **412**) in real time in order to control the hydrodynamics and uniform flow distribution, and hence the reaction chemistry, while also maintaining the solid support **424** packing inside the reactor **420**. For example, as the length of the oligomer increases, the load on the piston **220** will increase, and thus the piston **220** will be commanded to move up to maintain the hydrodynamics across the solid support **424**.
(67) Once the synthesis has been completed, the piston automatically moves up where the compression load on the piston reaches zero at the unloading stage **414**. At this stage, there may also be a limit switch sensor that can be triggered when the piston **220** reaches the defined unloading position. In various embodiments, each stage **400**, **402**, **404**, **408**, **412**, and **414** may be performed in any order as needed.
(68) The actuation or translation of the piston **220** may be carried out by one of several biasing mechanisms, as shown in FIGS. **6**B-**6**E; although a single, upper piston is illustrated here, in various embodiments these mechanisms could also be used on a lower piston. For example, the piston **220** may be biased via a mechanical source such as a linear actuator, as shown in FIG. **6**B. In another example, the piston **220** may be biased through pneumatic or hydraulic pressure source (via pressure port **432**), as shown in FIGS. **6**C-**6**D. The structure of the piston **220** may be decided, in part, by the biasing mechanism. For example, while a piston **220** with a contiguously rigid stem **241** can be used for reactors **420** utilizing either linearly actuated or pneumatic/hydraulic mechanisms (e.g., FIGS. **6**B, **6**D), a piston **220** biased via pneumatic or hydraulic mechanisms may be configured with a non-rigid stem **436**, such as a flexible tube that permits influent to enter the reactor **420**. In some embodiments, the reactor **420** includes a pressure chamber **440** that contains a biasing end **444** of the piston **220** (FIG. **6**E). Upon the addition of fluid into the pressure chamber **440**, the incoming fluid biases the piston **220** at the biasing end **444**, resulting in a translation of the piston **220** and a compression of the solid support **424**. In this configuration, the inlet **236** may be positioned at a site other than the top of the stem of the piston **220**. FIG. **6**F shows a

pneumatically/hydraulically-powered actuation system that may be used to move the piston **220** either up or down within the column. The system includes a pneumatic/pressure source and vent ports both above and below the piston end plate to provide on-demand movement of the piston either upward or downward. In various embodiments, the actuator may be configured as a lead screw actuator that includes a motor, a gear box, a lead screw, and a leadscrew nut. In other embodiments, the reactor **420** and/or reactor system **242** may include other types of actuators including but not limited to belt-drive and rack-and-pinion actuators. While FIGS. **6**B-**6**D show a single piston, these mechanisms can also be applied to double dynamic pistons (e.g., FIG. **2**).
(69) In certain embodiments, the piston **220** may further include a spacer between the fluid distribution plate and diffuser. The spacer creates a space between the diffuser and the fluid distribution plate, which may improve uniform flow distribution in the reactor **420** and may reduce the accumulation of by-products. The reactor **420** may also include a second fluid distribution plate, a second diffuser, a second spacer, and a second piston associated with the lower end of the chamber.
(70) In some embodiments a reactor may have a gas entry port on the effluent side of the reactor (e.g., on the bottom of the chamber), and a gas exit port on the influent side of the reactor (e.g., on the top of the chamber). Using the gas entry and exit port the reactor minimizes incomplete wash-out of reagents between synthesis steps/cycles via flushing the solid support with an inert or nonreactive gas (e.g., helium or nitrogen) through the gas entry port and the gas exit port. In this manner, a reactor system using the reactor may be configured as a hybrid-synthesis system with on-demand transitions between a packed bed and a gas-stirred fluidized bed, or between flow-through and batch beds/columns.
(71) In some embodiments, the piston may include a modular piston head assembly containing a diffuser that can be disassembled for cleaning and/or changing of the diffuser, as shown in FIGS. **24-26**. In accordance with one or more embodiments of the disclosure, the modular piston head assembly fits within the vessel. The diffuser fits within the modular piston head assembly where it acts to enhance flow distribution of influent through the solid support.
(72) In some embodiments, the reactor may include a flow distributor **600** within the upper piston **220**, as shown in FIGS. **7**A-**7**D, in accordance with one or more embodiments of the disclosure. The flow distributor **600** is intended to improve homologous distribution of influent over the whole diameter of the solid support **424**. For example, the flow distributor **600** may be configured to generate an axial flow of influent entering the reactor **300**. The flow distributor **600** may also be configured to generate other types of flow including but not limited to radial flow, laminar flow, or a hybrid flow that combines two or more of axial flow, radial flow, and laminar flow. In some embodiments, the reactor **300** may include a perforated inlet distributor **604** configured to generate uniform flow distribution as the influent moves through the reactor **300**, as shown in FIG. **7**D.
(73) FIG. **7**A shows a diagram of a flow distributor **600**, where arrows represent flow paths through the distributor. In some embodiments, the flow distributor **600** may include a grid for regulating flow, where the grid includes a radially asymmetric distribution of pore sizes having smaller pore sizes at the center (where flow velocity is higher) and where the pore size increases radially (as flow velocity is lower away from center) (see FIGS. **7**A, **7**B). FIG. **7**B shows a micrograph of a 3D printed mesh that can be used as part of a flow distributor such as in FIG. **7**A.
(74) FIG. **7**C shows a diagram of flow through a flow distributor (where the central axis of the chamber is oriented horizontally), with the front of flowing material being represented as an intense (red) band. The four examples of the flow presented in FIG. **7**C represent flow patterns at different flow quotients, or FQ, where FQ=(radial flow/axial flow), and where the first panel (left to right) is FQ=1, the second panel is FQ=2, the third panel is FQ=20, and the fourth panel is FQ=100. As demonstrated by the diagrams of FIG. **7**C, increasing radial flow relative to axial flow leads to a more uniform flow front and thus more evenly distributed flow of material through the column.
(75) In some embodiments, the upper piston **220** within the reactor **300** may include a nonuniform

distributor **606** disposed between the distribution plate **325** and the inside surface **474** of the first piston **220** (FIG. **7**A). The nonuniform distributor **606** may include a nonuniform distribution of pores, or nonuniform distributions of pore sizes, that increase the uniformity of influent as it passes through the solid support **424**. For example, the nonuniform distributor **604** may include narrow pore regions **608** at the center of the diameter of the reactor **300** to prevent an overabundance of influent from the central inlet **236** passing through the center of the diameter of the reactor **300** (FIG. **7**A, **7**B). In another example, the nonuniform distributor **604** may include narrow pore regions **608** near the perimeter of the vessel wall **208** to prevent channeling of influent between the vessel wall **208** and the solid support **424**. The nonuniform distributor **604** may include pore regions of multiple pore sizes (e.g., medium pore regions **612** and wide pore regions **616**, FIG. **7**A). For example, the nonuniform distributor **604** may include 2, 3, 4, 5, 10, or 20 or more pore regions of different pore sizes. In another example, the nonuniform distributor **604** may include a pore region having a gradient of pore sizes. An image of the surface nonuniform distributor **604** is shown in FIG. **7**B.

(76) In embodiments, the first piston may include a diffuser stack that includes diffusers of different pore sizes or pore densities. The integration of a diffuser stack with diffusers of large to small pore sizes minimizes diffuser/distributor fouling/clogging by capturing large particles at top layer as well as enhancing uniform flow distribution. The diffusers may be configured with any size of pore. For example, a diffuser with the smallest pore size may have a pore size of approximately 20 μm. The diffuser stack may include any number of diffusers including but limited to 2, 3, 4, 5, or 10 or more diffusers.

(77) In embodiments, the reactor or reactor system may include a diffuser housing that houses one or more diffusers and/or a diffuser stack. The diffuser housing may be coupled to the inlet via a first line. The diffuser housing may act as a pre-filter or guard column and may include multiple diffuser layers from large pore size to small pore size. The diffuser housing may act to minimize diffuser fouling and clogging and may enhance uniform flow distribution. The modular nature of the diffuser housing allows the diffuser housing to be cleaned or replaced independent of the reactor. The diffuser housing may further include crossflow ports that can be used for rinsing/cleaning.

(78) In embodiments, the reactor **300** or reactor system **242** includes a degasser housing **632** fluidly coupled to the vessel **204** (FIG. **8**). The degasser housing **632** functions as an in-line degasser, removing gas bubbles through the action of a porous membrane within the degasser housing **632**. The degasser housing **632** may be integrated before the inlet **236** and/or after the outlet **237** relative to fluid flow. Degassing of the fluid before it is disposed within the reactor **300** or reactor system **242** prevents gas bubble formation in pump heads or the sensors **244***a-b* (e.g., such as in an ultraviolet cell). The degasser housing **632** may be fluidly coupled to the diffuser housing **620** via a second line **636**.

(79) FIG. **9**A shows the configuration of a radial flow column (e.g., a concentric column) according to various embodiments. A radial flow column provides more homogeneous distribution of the fluid over the whole diameter of the packed bed at higher flow rates and lower pressure drop compared to axial flow columns. FIG. **9**A is a drawing illustrating a side view of a cross-section of a reactor **700** that includes inlets **236** that run along the length of the vessel **204**. The inlets **236***a-b* are separated from the solid support **424** by one or more distribution plates **325**. The inlets **236**, solid support **424**, and distribution plates **325** may be arranged as concentric circles. In this manner, influent may then enter into an outer “ring” of the reactor **700**, then pass through a distribution plate **325**, resembling a hollow cylinder, to the solid support **424**. The influent then moves through the solid support **424** to an outlet chamber **704** and is released via the outlet **237**. The concentric column-based reactor **700** may provide more homogenous distribution of fluid over the whole diameter of the solid support **424** (e.g., packed column) at higher flow rates and lower pressure drops compared to axial flow columns.

(80) In some embodiments, the reactor **700** includes a single inlet **236**, such the inlet **236** at the stem of the first piston **220**, as shown in FIG. **9**B. Inlet **236** is fluidly coupled to a system of fluid paths **708** (e.g., a branch-like flow distributor) within the solid support **424**. Fluid entering the fluid paths **708** then enters into the solid support **424**. The fluid then exits the solid support **424** by passing through the redistribution plate **325** and into the outlet chamber **704**. The fluid then leaves the reactor **700** at the outlet **237**. The redistribution plate **325** and the outlet chamber **704** surround the fluid paths **708** and the solid support **424** as concentric rings. In this manner, the reactor **700** may provide a combined axial and radial flow, providing homogeneous distribution of fluid over the whole diameter of the solid support **424**. The configuration of FIG. **9**B provides more homogeneous distribution of the fluid over the whole diameter of the packed bed in a combined axial and radial flow configuration (e.g., branch-like flow distributor).
(81) In some embodiments, the fluid paths **708** may be configured as an elongated and/or spiraled tube **712** having multiple apertures for the transfer of influent into the solid support **424** (e.g., analogous to a “soaker hose” for watering plants), as shown in FIG. **9**C. The configuration of FIG. **9**C provides more homogeneous distribution of the fluid over the whole volume of the packed bed using the spiral/spring-shaped perforated pipe. As a result, different layers of the solid support bed experience similar concentration of the reagents at the same time.
(82) In some embodiments, the reactor **700** includes a convective flow chamber **716**, as shown in FIG. **9**D. For example, a convection chamber **700** may be disposed at an end of the vessel **204** opposite the inlet **236**. Influent entering the inlet **236** travels through an inlet line **720** fluidly coupled to the inlet **236** and the convection chamber **716**. Upon entering the convection chamber **716** the influent is mixed via convection. The mixing enhances flow distribution and minimizes reagent channeling (e.g., preferential flow) by jetting the influent to the bottom surface **724** of the vessel **204**, which introduces more turbulent flow to the solid support **424** from bottom to top. The mixed influent then passes through the solid support **424** and out of the outlet **237**. The convection flow generated from the flow of fluid through the convection chamber **716** maximizes the wetted surface to volume ratio of the solid support **424** and minimizes drops in pressure. The convection chamber **716** also improves the contact surface of the solid support **424**/resin, enhances mixing, mass transfer, and reactor efficiency. The configuration of FIG. **9**D may help facilitate convective flow and maximize the wetted surface-to-volume ratio and minimize pressure drop. This configuration may also improve the CPGs/resin contact area and enhance mixing, mass transfer, and/or efficiency. Finally, the configuration may also enhance flow distribution and minimize reagents' channeling (preferential flow) by jetting the influent to the bottom surface of the column, which introduces more turbulent flow to the packed bed from its bottom to top.
(83) FIG. **10**A is a side cutaway view of a reactor **800** that includes a rotational blade, or in-line mixer, **804**, in accordance with one or more embodiments of the disclosure. The in-line mixer **804** mixes incoming influent coming in from the inlet **236**, improving the distribution of the influent through the solid support **424**. The in-line mixer **804** may include movable elements, such as a fan blade **808**, which introduces rotational flow to the solid support **424**. The in-line mixer **804** may be passively rotated, with rotation caused by the movement of fluid through the reactor **800**, or active, with rotation of the fan blade **808** powered by a small motor. In embodiments, the reactor **800** may further include one or more paddles **812** that rotate within the reactor **800**, mixing the solid support **424** as fluid transfers through the solid support **424** as shown in FIG. **10**B. The paddles **812** may be coupled to the in-line mixer **804**. The paddles **812** may rotate passively due to fluid flow, or actively via a small motor. The embodiments of FIGS. **10**A and **10**B facilitate convective flow and/or mixing of components, for example by maximizing the wetted surface-to-volume ratio and minimizing pressure drop, and/or improve the CPGs' contact area, thereby enhancing one or more of mixing, mass transfer, and/or efficiency. In addition, the embodiment of FIG. **10**A enhances flow distribution and minimizes the reagents' channeling as well as diffuser/distributor plate's surface fouling by placing a rotating blade under the diffuser/distributor plate, which introduces a rotational

flow to the packed bed. Further, the embodiment of FIG. **10**B provides hybrid synthesis systems, namely an on-demand transition between packed bed and mechanically-stirred fluidized bed. Finally, the embodiment of FIG. **10**B enhances flow distribution and mixing and minimizes reagents' channeling as well as diffuser/distributor plate's surface fouling by placing rotating blades inside the column, leading to agitation and repositioning of the packed resin/CPG.

(84) FIG. **11**A is a drawing illustrating a side cutaway view of a reactor **900** that includes an ultrasonic ring **904** enclosed around a section of the reactor vessel wall **208**, in accordance with one or more embodiments of the disclosure. The ultrasonic ring **904** emits an ultrasonic pulse towards the reactor **900**, which introduces agitation to the reactor **900** in the form of mechanical waves **908**. The ultrasonic ring **904** may be controlled by a signal generator **912**, which may itself be controlled by the controller **256**. The mechanical waves **908** enhance mixing within the vessel **204**. A top plan view of the reactor **900** is shown in FIG. **11**B. In some embodiments, the diffuser **470**, diffusion plate **325**, first piston **220**, or other component of the reactor **900** contains piezoelectric materials, such as polyvinylidene fluoride (PVDF) that vibrate when acted upon by the signal generator **912**. Beyond mixing, the vibrations may also prevent deposition of foulant on the surfaces of the reactor components. The embodiment of FIGS. **11**A and **11**B demonstrates how ultrasound-assisted synthesis may enhance the reactions and mixing, and hence enhance the overall process of synthesis, via introducing agitation (mechanical waves) to the column.

(85) FIG. **12** is a drawing illustrating a side cutaway view of a reactor **1000** that includes multiple distribution plates **325** *a-d* that define multiple sections **1004***a-c* packed with solid support **424**. The solid support **424** may enter each section **1004***a-c* with respective loading ports **238***a-c* and unloading ports **239***a-c*. The embodiment of FIG. **12** thus provides a multilayer bed/column which helps to minimize the flow channeling (preferential flow) which may occur in tall and large diameter beds.

(86) In some embodiments a reactor may be configured with a vessel having a high reactor width (e.g., or diameter) to reactor length ratio (with the solid support also having a high width to length ratio), providing a high aspect ratio of the width to the length/height of the reactor compared to other reactor embodiments disclosed herein. The reactor may include one or more distributor plates, flow distributors, or diffusers disposed within the vessel or top piston. The reactor may include any reactor width to reactor length aspect ratio including but not limited to a 1:1, 2:1, 3:1, 4:1, 5:1, 10:1, 15:1, 20:1 or higher ratios, or approximate ratios therein. For example, the reactor may be configured with a width of 15 cm and a length of 1 cm (e.g., a 15:1 ratio). The high reactor width to reactor length (W:H) reactors generally operate with a lower pressure drop when fluid flows axially across the packed bed than low W:H reactors. The reactor may also include inlets and gas entry ports that are combined or separated, as well as outlets and gas exit ports that are combined or separated. For example, the reactor may include a combined outlet and gas exit port controlled by one or more valves that control movement of effluent out of the reactor and the transfer of gas in and/or out of the reactor. The reactor may include different combinations of distributor plates, diffusers, and flow distributors, and may also include a more complex combined gas exit valve and outlet.

(87) In some embodiments, the reactor is configured such that the influent may fill a transient reagent chamber embedded within the piston and bounded by distributor plates, diffusers, or flow distributors. For example, gas may enter through a central gas entry port, with influent entering through an adjacent inlet. Gas may exit through an adjacent vent, while the effluent exits on the opposite side of the solid support. In this manner, pressure entering the transient reagent chamber is uniformly applied from the top of the reactor and uniformly pushes the influent through the solid support with a low pressure drop. In some cases, influent may initially fill the transient reagent chamber located below the solid support. In this configuration, one or both of the pistons may have a relatively flat piston head (e.g., the piston not enclosing the transient reagent chamber or the other piston not enclosing the transient reagent chamber). In this manner, pressure entering the transient

reagent chamber is uniformly applied from the bottom of the reactor and uniformly pushes up the influent through the solid support with a low pressure drop. In this configuration, the volume of the solid support may be fixed between the surrounding flow distributors or diffusers. In some embodiments where the reactor has a second piston, the second piston may also have a relatively flat piston head.

(88) In some embodiments, the transient reagent chamber is located within the second piston. For example, influent may enter the transient reagent chamber from the bottom of the reactor. The solid support is then exposed to influent only when pressure from incoming gas is uniformly applied to uniformly push the influent through the solid support. In some embodiments, the reactor includes two transient reagent chambers, one on each side of the reactor. The two transient chambers may be encompassed within, or be adjacent to, the first piston and/or second piston. For example, influent may first fill one or the other transient reagent chambers then gas may be uniformly applied from either end of the reactor to uniformly push the reagent through the solid support.

(89) In some embodiments, reactor **1100** includes one or more sensors **244** (e.g., a top sensor **244***a* and a bottom sensor **244***b*) that detect and/or measure one or more characteristics of the reactor **1100**, such as pressure within the vessel **204** or load applied to the pistons, as shown in FIG. **13**. Influent enters top or bottom transient reagent chambers **1116***a-b*, and either the top piston **220** or the bottom piston **232** moves, or inert gas enters, the top or bottom transient reagent chambers **1116***a-b* to uniformly push the influent through the solid support **424**. The sensors **244** may be located anywhere within, or adjacent to, the reactor **1100**. The sensors **244***a-b* may further include any type of sensor including but not limited to temperature sensors, pH sensors, conductivity sensors, load cells, strain gauges, and/or UV sensors. In the embodiment of FIG. **13**, the reagents may first fill the top/bottom transient reagent chambers, after which either the bottom dynamic flat (without embedded transient chamber) piston moves upward, or inert gas is added to the transient chamber embedded in the top piston to uniformly push the reagent through the resin with a lower pressure drop. Although the embodiment of FIG. **13** shows one particular example of where the pressure sensors may be located, in other embodiments one or more sensors could be placed in other locations and in addition to a pressure sensor the sensors could include one or more of sensors for temperature, pH, conductivity, UV, etc., which may also be integrated with the column.

(90) FIGS. **14**A-**14**B are drawings illustrating a side cutaway view of a reactor **1200** containing a solid support **424** that itself includes a plurality of solid support units **1204** (e.g., CPG/resin+oligomer). The solid support units **1204** may include any type of solid support material including but not limited CPG or resin. The solid support units **1204** may include, be crosslinked to, or otherwise be coupled to, a linked substituent including but not limited to an oligomer, such as an oligonucleotide. The solid support **424** may further include a support spacer **1208** (FIG. **14**B), an inert component that does not bind to influent components, but rather introduces extra space or voids within the solid support **424**. For example, for a solid support **424** containing solid support units **1204** that include a linked oligomer, the support spacer **1208** may increase the ability of the oligomer to extend outward from the solid support unit (e.g., increasing the functional length of oligomer). This support spacer **1208** improves the functional length of the oligomer with minimal pressure elevation within the column. The support spacer **1208** may take on any physical form, such as a spherical or other shaped bead or an elongated shape such as a grain shape. The support spacer **1208** may include any type of material including but not limited to glass and quartz microbeads, polystyrene, and CPG. For example, the support spacer **1208** may include dry polystyrene or polystyrene pre-swelled in a solvent such as toluene. In the embodiment of FIG. **14**B, the addition of inert filling/spacer materials introduces extra space/void to permit the oligomer to increase in length with minimum pressure elevation within the column. Other solid supports including dry polystyrene (PS) or pre-swelled PS in a solvent (e.g., toluene) can also be used as resins.

(91) In some embodiments, the vessel **204** and/or the influent may be heated or cooled to control

(e.g., increase or decrease) chemical reaction rates, as shown in reactor **1300** of FIGS. **15**A-**15**D, according to one or more embodiments of the disclosure. For example, the reactor **1300** in FIG. **15**A is shown having a fluid entry port **1301** that allows cooled or heated fluid or inert gases to enter the vessel, which raises or lowers the temperature of the solid support **424** within the vessel. The fluid may then exit the vessel **204** via the fluid exit port **1302** or other opening. In another embodiment, the reactor **1300** may include a jacket **1304** that warms or cools the vessel **204**, as shown in FIG. **15**B. For example, the jacket **1304** may receive warmed or cooled fluid via the fluid entry port **1301**, which warm the vessel **204** during the transit of the fluid through the fluid exit port **1302**. In some embodiments, the reactor **1300** includes a heat exchanger **1308** that increases or decreases the temperature of the influent before the influent enters the vessel **204**, as shown in FIG. **15**C. In some embodiments, the reactor **1300** includes a heat exchange housing **1312** that heats and/or cools the vessel **204** directly (e.g., without circulating fluid or influent) as shown in FIG. **15**D. The heat exchange housing **1312** may use any type of heating and/or cooling technology including but not limited to microwave heating, inductive heating, and Peltier heating/cooling. In general, a heat exchanger may be integrated into the system in cases where pre-heating or cooling of the reagents is required to increase the chemical reaction rates, hence reducing total synthesis time. A heat exchanger may be integrated into the system to provide heat using microwaves, radiative, or inductive mechanisms of heat transfer (e.g., an element may be placed inside the column which can be actuated externally), and heat exchangers may also be used to readily create a uniform temperature across the column.

(92) FIGS. **16**A and **16**B are perspective and near-plan views of a degasser module **640**, respectively, in accordance with one or more embodiments of the disclosure. The degasser module **640**, which may degas as well as heat incoming fluids (e.g., instead of or in addition to the fluid warming mechanisms disclosed in FIG. **15**), may be disposed within the degasser housing **632** or may itself include a housing element. The degasser module further includes a gas block **642** that houses a gas/bubble outlet **644** that allows gas to escape the degasser module **640**. The degasser module **640** further includes a gas-permeable porous membrane **634** and a liquid block **646** that includes a degasser inlet **648** and a degasser outlet **650** as well as internal channels that allow liquid to be exposed to the gas-permeable porous membrane **634**. The degasser module **640** may further include a heater element **652** that may be controlled by a programmable heat source. FIGS. **16**C and **16**D are exploded views of the degasser module viewed from different angles. The exploded view in FIG. **16**D exposes the fluid channel interface **654** which permits fluid from within the fluid block **646** to contact the gas-permeable porous membrane **634**. The exploded view in FIG. **16**C exposes the gas channel interface **656** that permits gas to pass into the gas block **642** from the gas-permeable porous membrane **634**. In embodiments of the device of FIGS. **16**A-**16**D, reagents (with or without bubbles) enter the device via the inlet port. While passing through the device and above the porous gas-permeable membrane, bubbles are removed from the reagents. The temperature of the reagents can also be controlled via the programmable heat source while passing through the integrated device.

(93) FIG. **17** is a drawing illustrating a side cutaway view of a reactor **1400** having a multitude of sampling ports **1404***a-d* (in-process sampling ports), in accordance with one or more embodiments of the disclosure. Each sampling port allows a user to obtain a sample of influent, effluent, solid support (e.g., resin) or other reactor component for quality assessment. The reactor **1400** may have any number of any type of sampling port **1404***a-d*. For example, the reactor **1400** may include an inlet sampling port **1404***a* for sampling influent. In another example, the reactor **1400** may include an outlet sampling port **1404***b* for sampling effluent. In yet another example, the reactor **1400** may include one or more resin sampling ports **1404***c-d* to determine the quality of the resin, or the binding of materials to the resin. The embodiment of FIG. **17** may facilitate in-line and in-column quality control (QC) sampling, where resin and/or fluid sampling may be performed at different stages of a synthesis process for QC or other purposes.

(94) FIG. **18**A is a drawing illustrating a side cutaway view of a reactor **1500** having a structured support **1504** within the vessel **204**, in accordance with one or more embodiments of the disclosure. FIG. **18**B shows a photograph of the structured support from the top and side and a diagram of the side view of the support filled with resin. The structured support **1504** features a scaffold or framing with pockets that are filled with resins or other solid support materials. Reactors **1500** that include the structured support **1504** may have lower pressure drops, higher flow rates, and enhanced flow distribution as compared to vessels **204** that do not contain structured supports **1504**. The structured support may also effectively allow synthesis reactions to occur within the reactor **1500** using relatively small volumes of reagents (e.g., influent and/or solid support). The structured support **1504** may bind, or not bind, components within the influent. The structured support **1504** may be configured as a self-supporting structure, such as a cylinder that can be inserted into the vessel **204**, or as a structure that is applied as a multitude of layers into the vessel **204**. In various embodiments, structured packed resins may include flexible structured scaffolds/frames that are filled with resins which may be used inside the column. Using a structured support provides one or more of: a lower pressure drop, higher flow rates, and/or enhanced flow distribution. The empty spaces inside the scaffolds may be filled with functionalized resins that are confined in a microenvironment for effective reactions while using much lower volumes of reagents.
(95) FIG. **19**A is a perspective and partially transparent view of reactor **1600**, in accordance with one or more embodiments of the disclosure. The reactor **1600** includes a resin-free vessel **204** that includes an array or stack of plates **1604** (e.g., flat plates) or hollow tubes with functionalized surfaces **1608** containing linkers, or other chemical moieties. The plates **1604** or tubes may be placed in a radial (e.g., as shown in FIG. **19**A) or an axial/parallel configuration within the vessel **204** (FIG. **19**B). Oligomers **1612** may then be synthesized directly from the surface **1608** of the plates **1604** as influent flows pass via convection. The plates **1604** or tubes may also include filaments or polymer brushes to provide a surface **1608** for synthesizing the oligomers **1612**. The reactor **1600** may have lower pressure drop and greater flow characteristics than other vessel designs.
(96) In some embodiments, configurations such as those shown in FIGS. **19**A and **19**B provide for a convection-dominated flow by providing a resin free oligo synthesizer system; in contrast, resin-based oligo synthesizers are based on diffusion-dominated flow. An array or stack of flat plates (e.g., as shown in FIGS. **19**A, **19**B) or hollow tubes with functionalized surfaces (which include monolayers and/or linkers attached to the surfaces) may be positioned in an axial (parallel, FIG. **19**B) or radial (shown in FIG. **19**A) configuration inside the synthesizing column. In the configurations of FIGS. **19**A and **19**B, oligomers will grow directly on the surfaces of the flat plates, as opposed to growing inside resin pores when CPG or other (porous) resins are used. In configurations such as FIGS. **19**A and **19**B, polymer brushes may also be used instead of or in addition to monolayers to boost the yield of synthesis. Multiplex synthesis is facilitated using these column designs in part due to the columns' low pressure drop and similar flow characteristics among columns.
(97) In various embodiments a reactor may include a resin-free vessel which itself includes a porous membrane with functionalized groups attached to the surface of the membrane. The porous membrane may be configured as a flat plate or as a series of hollow tubes. Influent is then driven by a pressure difference across the porous membrane or by a specific flow rate. In this manner, the influent, and influent components, are transported by convection of the influent through the pores of the porous membrane. Multiplex synthesis is possible through reactor due to the low pressure drop and flow characteristics of the reactor.
(98) It should be understood that any of the reactors disclosed herein may include one or more, or all, components of any other reactor, and vice-versa. Any of the disclosed reactor systems may utilize any reactor, and the reactor may be of any size and made of any material including but not

limited to glass, plastic, or stainless steel. The reactor may have any number of components described herein. For example, a reactor shown with a single first piston **220** may also include a second piston **232**, and a reactor shown with both a first piston **220** and a second piston **232** may be implemented within the reactor system **242** with only the single first piston **220**, or the single second piston **232**.

(99) FIGS. **20-22** are flow schemes **1800**, **1900**, **1950** for synthesis that include the reactor **300**, in accordance with one or more embodiments of the disclosure. The flow schemes **1800**, **1900**, **1950** may include, or be included within, any reactor system **242**. The flow schemes **1800**, **1900**, **1950** may also utilize any reactor or number of reactors disclosed herein. The flow schemes **1800**, **1900**, **1950** may include additional componentry and different flow paths to achieve polymer synthesis using the reactors.

(100) Flow scheme **1800** includes reactors **200***a-d* that include, are adjacent to, or are in-line with, one or more sensors **244***a-b* (e.g., pressure sensors). Other sensors **244***c-e* may be included within the flow scheme **1800** and/or interconnected to the reactors **200***a-d* including but not limited to a pH sensor **244***c*, a conduction sensor **244***d*, and a UV sensor **244***e*. The flow scheme **1800** includes a master vent **1804** coupled to reactor vents **340**. The flow scheme **1800** includes a plurality of valves **1112***a-d* that control the flow of gas and fluids (e.g., influent, effluent). The flow scheme **1800** further includes a series of reagent storage units **1808***a-d* that store reagents (e.g., nucleotide solutions, wash solutions, reaction solutions) that are coupled to the rest of the flow scheme **1800** via unit valves **1812***a-d*. The flow scheme **1800** may also include gas storage units **1816** coupled to the rest of the flow scheme **1800** via a gas valve **1820**.

(101) The flow scheme **1800** may further include one or more pumps **1824***a-b* that drive fluids through the flow scheme **1800**. The flow scheme may further include fluid control units **1826***a-b*. The fluid control units **1826***a-b* may control the temperature of the fluid (e.g., via a heat exchanger) or may degas the fluid via a degasser. The flow scheme **1800** may also include a gas control unit **1828** that controls the temperature of the gas (e.g., via a heat exchanger) or may remove moisture from the gas via a drying filter or other drying device.

(102) The flow scheme **1800** may further include a waste unit **1832** controlled via a waste valve **1836** and a waste flow restrictor/back pressure regulator **1840**. The flow scheme may further include a collection unit **1844** for collecting finished product that is controlled via a collection valve **1848** and a collection flow restrictor/back pressure regulator **1852**. The flow scheme may further include a series of unit valves **1856***a-c* that control fluid movement between the waste unit **1832**, the collection unit **1844**, the reagent units **1812***a-d*, and the gas storage unit **1816**. The flow schemes **1800**, **1900**, and **1950** may each also include an optional pressure sensor **1821**, relief/control valve **1822**, and relief/control line **1823** that directs fluid toward the waste unit **1832** if the flow schemes **1800**, **1900**, **1950** are exposed to high pressure or other aberrant circumstances.

(103) Flow scheme **1800** is designed to run at least four reactors **200** in parallel. Furthermore, the flow scheme may include additional valves **1112***a-d* and in-line sensors **244**, making it possible for the flow scheme **1800** to run all reactors **200** in parallel and independent of each other (e.g., as a multiplexed reactor set). Reagents and other fluids may be delivered to the reactors **200***a-d* from the top or bottom of the reactor **300***a-d*. The flow scheme **1800** may also include other sensors **244**, such as temperature sensors.

(104) Flow scheme **1900**, which is similar to flow scheme **1800**, further includes a pre-reactor chamber **1904** placed above or below the reactors **200***a-d*. The pre-reactor chamber **1904** may be pre-filled with reagents via the one or more pumps **1824***a-b*. The pre-reactor chamber **1904** may also be pressured by inert gas from the gas storage unit **1816** that pushes reagents into the reactors **200***a-d* at faster flow rates. Flow scheme **1950**, which is similar to flow scheme **1900**, includes rotary/switch/selector valves **2054***a-b* in place of the on/off valves **1112***a-d* that are seen on either side of the reactors **200***a-d* in flow schemes **1800**, **1900**.

(105) FIG. **23** is a flow diagram illustrating a method **2000** for transferring influent through a

reactor, in accordance with one or more embodiments of the disclosure. The transfer of influent through a reactor may include but not be limited to the addition of a wash solution through a solid support **424**, the addition of reagents to the solid support **424** during synthesis of an oligomer, or the addition of reagents to the solid support during the cleaving and/or purification of a compound from a mixture.

(106) In embodiments, the method **2000** includes a step **2004** of receiving an influent through an inlet **236** of a reactor **300**. For example, the influent may be transferred through the inlet **236** via one or more pumps **1824** operatively coupled to the inlet **236** via tubing.

(107) In embodiments, the method **2000** includes a step **2008** of driving the influent through the solid support **424** of the reactor **300**. The reactor may include any method or technology to drive, bias, or force, the influent through the solid support **424** including but not limited to: translating a first piston **220** to push or bias the influent into the solid support; using pressurized inert gas to push the influent into the solid support **424**; or forcing the influent through the solid support **424** via pumps operating at specific flow rates or flow pressures.

(108) In embodiments, the method **2000** includes a step **2012** of measuring at least one characteristic of the influent as the influent flows through the solid support **424** and/or at least one characteristic of the effluent as the effluent leaves the reactor **300** and/or at least one characteristic of the flow inside the reactor **300**. The at least one characteristic may include but not be limited to one or more load readings, one or more pressure readings, one or more UV readings, one or more temperature readings, one or more pH readings, one or more conductivity readings, one or more influent flow readings, or any other type of influent characteristic.

(109) In embodiments, the method **2000** includes a step **2016** of changing a parameter of the reactor **300** or influent and/or effluent based on a measurement of the at least one characteristic. For example, the reactor **300** may change one or more parameters including but not limited to a first piston position, a second piston position, the temperature of the reactor **300** (e.g., via heat exchangers **1308**), the vibration of the reactor **300** (e.g., via signal generators **912**), influent pump speed, influent pressure limits, or paddle speed (e.g., via paddles **812**). The method **2000** further includes a step **2020** of releasing the effluent via the outlet **237** of the reactor **300**.

(110) FIG. **24** shows that in certain embodiments the modular flow distributor can include three main parts: distributor head, distributor cap, and filter. These parts can be made of different chemical resistant materials including, but not limited to, natural PEEK, stainless steel, and Teflon. The distributor head may include inlet/input ports for influent, cross flows, and/or sensors, and may also include branched outlets to more uniformly distribute the influent on and through the filter. Although the cross-sectional views in FIG. **24** show two branched outlets extending from the central outlet, in various embodiments any number of branches may extend outward (e.g., the assembled view in the upper right panel shows six openings in the face of the distributor head corresponding to six branches extending outward from the central outlet). FIG. **24** also shows a slot in the distributor head for an inner O-ring and a slot in the distributor cap for an outer O-ring. Although only one O-ring slot is shown in the distributor head and the distributor cap in FIG. **24**, in various embodiments either or both components can have any number of O-rings. In certain embodiments, a fixed amount of head space may be present between the face of the distributor head and the filter screen. In some embodiments, the head space may contain at least one of a filter, a porous material, a screen, or a reagent. Instead, in certain embodiments the porous outer piston may be replaced by a filter screen and the fixed head space may be directly designed into the inner piston head itself, similar to the assembly shown in FIG. **24**. In particular embodiments, a flow distributor may be used in one or both of the inlet and outlet side of the chamber to further improve uniform flow (e.g., FIG. **5**B). In general, the flow distributor includes a central channel (coupled to an inlet or an outlet) and a number of flow distribution channels extending radially outward from the central channel and terminating at the face of the flow distributor which faces the chamber; at the inlet side the flow distribution channels evenly distribute the influent across the reaction

chamber while at the outlet side the flow distribution channels evenly collect the effluent from the across the reaction chamber and channel this to the outlet, as shown in FIG. **5**B.

(111) FIG. **25** shows different parts of a modular flow distributor, which in some embodiments may be used with a large diameter column. The right-hand panels in FIG. **25** show that the distributor can be made of different materials (e.g., 3D printed SS-**316**L or CNC-machined PEEK) but is not limited to the materials shown in the figure, and furthermore can be manufactured via procedures including 3D printing, CNC machining, etc.

(112) FIG. **26** shows another design of a modular flow distributor which in various embodiments may be used for smaller diameter columns. Similar to the modular flow distributor for large diameter columns, the distributor of FIG. **26** includes three main parts: distributor head, distributor cap, and filter, and can be made of different chemical resistant materials including but not limited to natural PEEK, stainless steel, and Teflon. The distributor head of FIG. **26** includes one or more inlet ports for influent, cross flows, and/or sensors, and also may have branched outlets to more uniformly distribute the influent on and through the filter.

(113) FIG. **27** shows another design of a modular flow distributor which may include two top and bottom parts that may be clamped together using screws or other suitable fasteners. The top part may include the inlet ports for influent, sensors, etc., while the bottom part includes the filter which can be regularly cleaned by unclamping the top and bottom parts.

(114) FIG. **28** shows embodiments of stainless-steel wire mesh sintered filter discs which include multi-layer structures with different pore sizes at each layer. The pore size of the filter layer is selected such that it increases the ratio of lateral flow to axial flow, hence improving the uniform flow distribution, and it is also selected such that does not let the target particle size (smallest particles) pass through it.

(115) FIG. **29** shows a mechanically actuated system (linear actuator) to drive the piston (modular flow distributor) within the column in response to feedback received from sensors/detectors. The detailed view on the left shows an exploded view of the modular flow distributor assembly and the main drawing on the right shows how the modular flow distributor assembly (which is part of the piston) is coupled to a load cell by an actuator rod end coupler, where the load cell in turn is connected to the actuator. As disclosed herein the load cell is used to monitor reactions, and hence the real-time conditions inside the column, by measuring the force on one or both of the pistons.

(116) FIG. **30** shows data collected from a synthesis system at different stages during solid state oligonucleotide synthesis. In particular, FIG. **30** shows data from sensors (UV1, 504 nm; UV3, 260 nm; Cond (conductivity); and Pressure) which monitor the progress of various parameters during two cycles of deblocking and coupling (each cycle indicated by the horizontal arrows below the graph). The data in FIG. **30** shows that dynamic adjustment of the column height, by real-time positioning of the piston which is integrated with one or more load cells, affects the pressure distributions which represent the real-time conditions inside the column, which can be monitored by reading the loadcell set values. In this particular case, the piston moves down (hence reducing the column volume) during the deblocking step when the protection groups are removed from the bases, and conversely the piston moves up (hence increasing the column volume) when new bases are added to the previous bases. As a result, pressure remains almost constant during these two steps (see labels in FIG. **30**).

(117) FIG. **31** shows data collected from a synthesis system at different stages during oligonucleotide synthesis, comparing a static column (left) to a column that is dynamically adjusted during operation (right), with time in minutes being indicated on the x-axis; it should be noted that in a “static” column the volume of the reaction space is held fixed during the reaction which can result in pressure changes, whereas in a dynamic column the volume of the reaction space is adjusted dynamically which can adjust the pressure during the reactions (e.g., to hold the pressure at or near a constant level or to adjust to the pressure based on desired characteristics). FIG. **31** demonstrates how changes in pressure during synthesis lead to the dynamic adjustment of the

column's height via real-time motion of the piston in response to conditions inside the column which are read by sensors (e.g., load cells, pressure sensors etc.) compared to that of in a static column. As shown in the left panel of FIG. **31** (static column) the pressure increases over the course of the coupling step as the oligo length increases. However, as shown in the right panel of FIG. **31** (dynamic column) the pressure remains steady due to the feedback-driven adjustment of the volume of the column via movement of the piston(s).

(118) FIG. **32** shows a position of a piston during operation of a dynamic column. The trace in FIG. **32** presents real-time position changes (movement) of the piston inside a dynamic column during synthesis, where the time is indicated in hours on the x-axis. Depending on how the various parameters have been set, the volume of the column increases (i.e., there is a net change in piston position increases) over the long period of a synthesis, over which the length of the oligomer increases. As shown in FIG. **32**, the piston undergoes a regular pattern of increases and decreases in position corresponding to each cycle of extension of the oligonucleotides. As the overall length of the oligonucleotides grows from the start of the trace to the finish, there is a net upward movement in the position of the piston which represents the increase of volume of the solid state material (e.g., resin) inside the reaction chamber.

(119) FIGS. **33**A-**33**D show data indicating quality and quantity of a 100-bp oligo synthesized using several known systems (labeled “control”) compared to the same oligo prepared using the presently-disclosed procedures (labeled “current invention”). These data show the quality and quantity of a particular 100-base long oligomer synthesized using several known/commercially-available “control” column systems compared to columns which incorporate the presently-disclosed features.

(120) The performance of synthesis columns can be quantified in several ways. First, a high percentage of full-length oligonucleotides (i.e., full-length product or ‘flp,’ measured in percentage) is desired. The data presented in FIG. **33**A shows that the average full-length oligomer is higher when using the current invention compared to the control systems. Second, oligomer synthesis columns should produce a high crude yield so there is enough product to be purified. FIG. **33**B shows that the crude yield (measured in nmol) obtained using the disclosed synthesis systems is comparable to, or even greater than, the yield obtained with known/commercially-available “control” column systems. The graphs in FIGS. **33**C and **33**D show the amounts (indicated as percentages) of impurities with one missing base (FIG. **33**C, n−1) or one extra base (FIG. **33**D, n+1) for oligonucleotides synthesized using each of the known systems (controls) and the presently-disclosed procedures (current invention). The “spec limit” in each graph represents a maximum tolerable percentage of the particular impurity which can exist in the as-synthesized product before beginning further purification by HPLC or other post-processing techniques, where the spec limit can be determined in a number of ways (e.g., standard procedures, customer preference, etc.).

(121) FIGS. **34**A and **34**B show data obtained from the output of HPLC columns for oligomers prepared using dynamic (FIG. **34**A) or static (FIG. **34**B) pistons. In particular, FIGS. **34**A and **34**B show analytical HPLC histograms for oligomers prepared using synthesis columns with a dynamic (FIG. **34**A) or static (FIG. **34**B) piston, i.e., with a column having a variable/adjustable column height or a fixed column height, respectively. In each of the graphs of FIGS. **34**A and **34**B, the area under the main large peak represents the purity percentage of the synthesized oligomers, which are above 70-80% even for a relatively long oligomer such as one that is 80 bases long.

(122) Although inventive concepts have been described with reference to the embodiments illustrated in the attached drawing figures, equivalents may be employed and substitutions made herein without departing from the scope of the claims. Components illustrated and described herein are merely examples of a system/device and components that may be used to implement embodiments of the inventive concepts and may be replaced with other devices and components without departing from the scope of the claims. Furthermore, any dimensions, degrees, and/or

numerical ranges provided herein are to be understood as non-limiting examples unless otherwise specified in the claims.

Claims

1. A reactor system, comprising: a vessel configured to contain a solid support, the vessel comprising: a vessel wall defining a reaction chamber, the reaction chamber having a first end and a second end opposite the first end; a piston operatively arranged at the first end and configured to translate within the reaction chamber, the piston comprising a piston head coupled to a piston shaft, and the piston head comprising a plurality of crossflow ports disposed therein; a force measuring device attached to the piston shaft and configured to measure a load on the piston shaft; a piston driver coupled to the piston shaft; and a processor operably coupled to the force measuring device and the piston driver, the processor configured to: measure a load on the piston shaft using the force measuring device, adjust a position of the piston using the piston driver based on measuring the load on the piston shaft, and move fluid into and out of a space adjacent to the piston head using the plurality of crossflow ports.
2. The reactor system of claim 1, further comprising: an inlet disposed at the first end of the reaction chamber and configured to receive an influent; and an outlet disposed at the second end of the reaction chamber and configured to release an effluent.
3. The reactor system of claim 2, wherein the piston further comprises a flow distribution unit coupled to the piston shaft and disposed within the vessel adjacent the reaction chamber, wherein the flow distribution unit comprises a plurality of flow distribution channels, wherein a first end of each of the plurality of flow distribution channels is fluidly coupled to the inlet, and wherein a second end of each of the plurality of flow distribution channels is fluidly coupled to the reaction chamber.
4. The reactor system of claim 3, wherein the flow distribution unit comprises a modular flow distribution unit comprising a distributor head coupled to a distributor cap with a filter disposed therebetween, wherein the plurality of flow distribution channels is disposed within the distributor head.
5. The reactor system of claim 1, further comprising a pressure sensor operatively coupled to the processor and configured to measure a pressure within the vessel, wherein the processor is further configured to: obtain a measurement of the pressure within the vessel from the pressure sensor, and adjust the pressure within the vessel based on the measurement of the pressure comprising at least one of adjusting the position of the piston or actuating a pressure-control device comprising a valve.
6. The reactor system of claim 1, further comprising a solid support disposed within the reaction chamber, wherein the solid support is configured to support at least one of: oligonucleotide synthesis, pretreatment, cleavage, deprotection, or purification.
7. The reactor system of claim 1, wherein the piston is a first piston, and wherein the system further comprises: a second piston operatively arranged at the second end of the reaction chamber and configured to translate within the reaction chamber.
8. The reactor system of claim 1, wherein the vessel further comprises at least one of: a sampling port, a loading port configured to facilitate loading of the solid support into the reaction chamber, or an unloading port configured to facilitate unloading of the solid support out of the reaction chamber.
9. The reactor system of claim 1, wherein the force measuring device comprises at least one of a load cell or a strain gauge, and wherein the piston driver drives the piston using at least one of mechanical, hydrodynamic, or pneumatic force.
10. The reactor system of claim 1, wherein the load on the piston shaft measured by the force measuring device is proportional to a pressure within the reaction chamber.

11. The reactor system of claim 2, further comprising a sensor coupled to the inlet, wherein the sensor comprises at least one of a flow sensor, a pH sensor, a conductivity sensor, a thermal sensor, a spectrometer, a Raman sensor, or a mid-IR sensor.
12. The reactor system of claim 2, further comprising a fluid conditioner fluidly coupled to the inlet, wherein the fluid conditioner comprises at least one of a heat exchanger, a degasser, a de-bubbler, or a gas dryer.
13. The reactor system of claim 1, wherein the piston comprises a flow distribution plate coupled to the piston shaft and comprising a plurality of pores.
14. The reactor system of claim 13, wherein the plurality of pores of the flow distribution plate comprises different size openings, and wherein the different size openings of the plurality of pores are distributed in a radial pattern across the flow distribution plate.
15. The reactor system of claim 1, wherein the processor is further configured to: collect a plurality of real-time data based on obtaining a plurality of measurements of the load on the piston shaft using the force measuring device, wherein the plurality of real-time data indicates at least one of a reaction chemistry or flow dynamics within the reaction chamber, and adjust the position of the piston using the piston driver based on collecting the plurality of real-time data to adjust at least one of a system or operational parameter in real time.
16. A method of operating a reactor, the method comprising: providing a vessel configured to contain a solid support, the vessel comprising: a vessel wall defining a reaction chamber, the reaction chamber having a first end and a second end opposite the first end, a piston operatively arranged at the first end and configured to translate within the reaction chamber, the piston comprising a piston head coupled to a piston shaft, and the piston head comprising a plurality of crossflow ports disposed therein, a force measuring device attached to the piston shaft and configured to measure a load on the piston shaft, a piston driver coupled to the piston shaft, and a processor operably coupled to the force measuring device and the piston driver; measuring, using the processor, a load on the piston shaft using the force measuring device; adjusting, using the processor, a position of the piston using the piston driver based on measuring the load on the piston shaft; and moving, using the processor and via the plurality of crossflow ports, fluid into and out of a space adjacent to the piston head.
17. The method of claim 16, wherein providing a vessel further comprises: providing an inlet disposed at the first end of the reaction chamber and configured to receive an influent, and providing an outlet disposed at the second end of the reaction chamber and configured to release an effluent; and wherein the method further comprises at least one of: delivering the influent to the reaction chamber via the inlet, or obtaining the effluent from the reaction chamber via the outlet.
18. The method of claim 17, wherein providing the vessel further comprises: providing the force measuring device comprising at least one of a load cell or a strain gauge, and providing the piston driver wherein the piston driver drives the piston using at least one of mechanical, hydrodynamic, or pneumatic force.
19. The method of claim 16, wherein providing the vessel further comprises: providing a pressure sensor coupled to the reaction chamber; and wherein the method further comprises: measuring a pressure within the reaction chamber using the pressure sensor.
20. The method of claim 19, wherein, upon measuring an increase in pressure, adjusting the position of the piston further comprises: adjusting the position of the piston to increase a volume of the reaction chamber and reduce a pressure within the reaction chamber.
21. The method of claim 19, wherein providing the vessel further comprises: providing a sensor coupled to the inlet, wherein the sensor comprises at least one of a flow sensor, a pH sensor, a conductivity sensor, a thermal sensor, a spectrometer, a Raman sensor, or a mid-IR sensor; and wherein delivering the influent to the reaction chamber further comprises: measuring a property of the influent using the sensor.
22. The method of claim 18, wherein providing the vessel further comprises: providing a fluid

conditioner coupled to the inlet, wherein the fluid conditioner comprises at least one of a heat exchanger, a degasser, a de-bubbler, or a gas dryer; and wherein delivering the influent to the reaction chamber further comprises: adjusting a property of the influent using the fluid conditioner.

23. The method of claim 18, wherein providing the vessel further comprises: providing a flow distribution unit coupled to the piston shaft and disposed within the vessel adjacent the reaction chamber, wherein the flow distribution unit is fluidly coupled to the inlet; and wherein delivering the influent to the reaction chamber further comprises: delivering the influent to the reaction chamber using the flow distribution unit.

24. The method of claim 23, wherein providing the flow distribution unit further comprises: providing the flow distribution unit, wherein the flow distribution unit comprises a plurality of flow distribution channels, wherein a first end of each of the plurality of flow distribution channels is fluidly coupled to the inlet, and wherein a second end of each of the plurality of flow distribution channels is fluidly coupled to the reaction chamber; and wherein delivering the influent to the reaction chamber further comprises: delivering the influent to the reaction chamber using each of the plurality of flow distribution channels of the flow distribution unit.